

Operator's Digest

**SW11-615
R684
03/2013**

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About This Document

This publication supports TPS Network Release 684. TPS is the evolution of TDC 3000^X. The contents and procedures in this digest may vary somewhat from those applicable to your system because of continuing product evolution.

On any station that does not have a touchscreen, selection is accomplished by moving the cursor to a target and pressing the SELECT key in the keyboard or mouse pushbutton.

ATTENTION

With TPN R684, Enhanced Universal Control Network (EUCN) is introduced. In EUCN, the traditional coaxial cable system is no longer utilized and is replaced with a Fault Tolerant Ethernet (FTE) based communication network. Throughout the document, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms **EHPM**, **ENIM**, and **EUCN** respectively.

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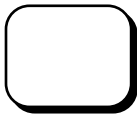
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LEGEND



Keyboard Button



Visible Touch Target



Invisible Touch Target



Target on System Menu



Indicates that the keyswitch must be in the Supervisor's position. The Supervisor's access level can be activated on the Universal Work Station by pressing keyboard keys [shift + PF1].



Indicates that the keyswitch must be in the Engineer's position. The Engineer's access level can be activated on the Universal Work Station by pressing keyboard key [PF1].

TPS ACRONYMS

Networks

Hiway	Data Hiway
LCN	Local Control Network
PIN	Plant Information Network
UCN	Universal Control Network
EUCN	Enhanced Universal Control Network

UCN Devices

APM	Advanced Process Manager
HPM	High Performance Process Manager
EHPM	Enhanced High Performance Process Manager
LM	Logic Manager
PM	Process Manager
SM	Safety Manager
XPM	Process Manager Family (PM, APM, HPM, and EHPM)

LCN Devices

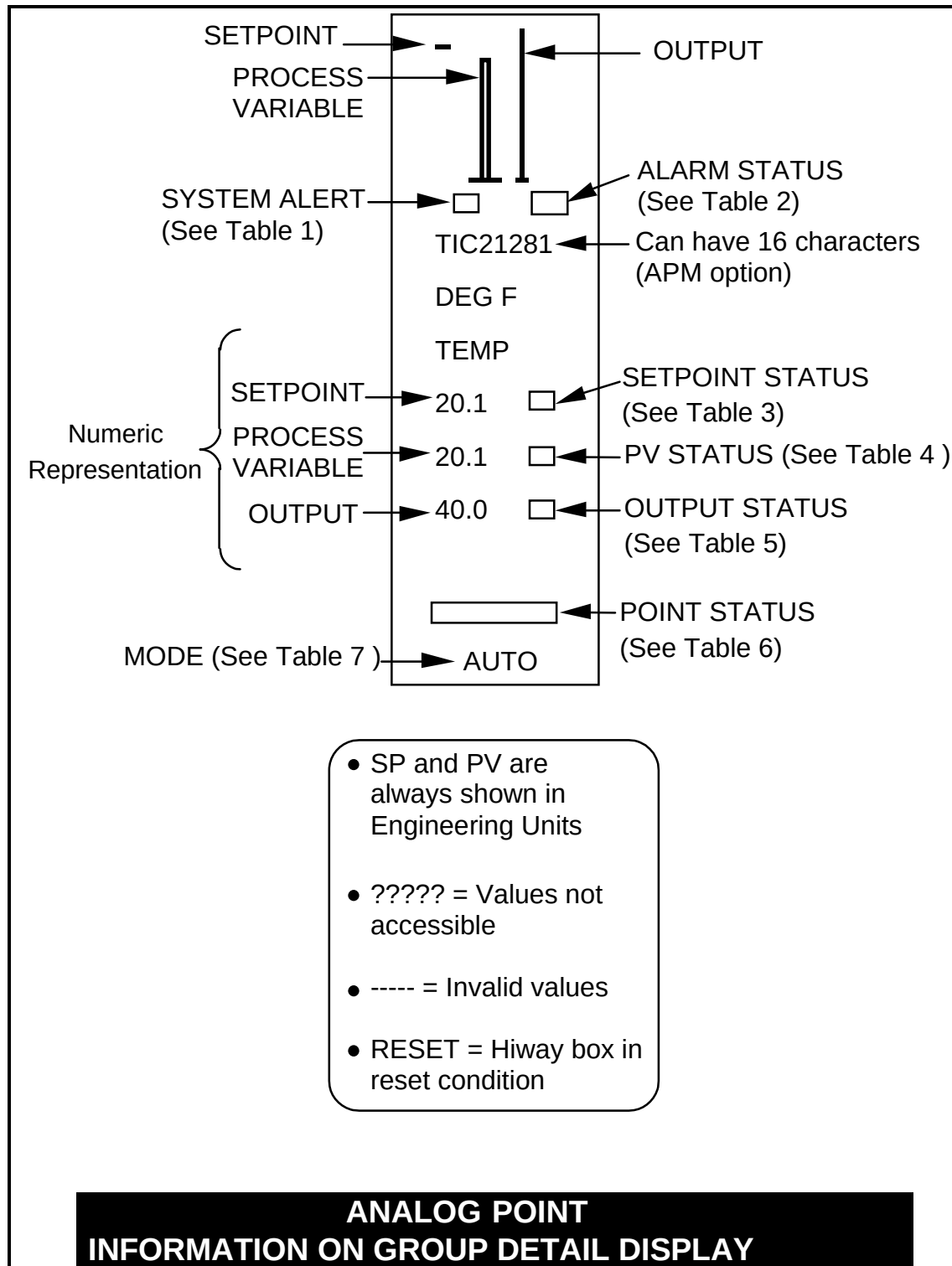
AM	Application Module
ARM	Archive Replay Module
CG	Computer Gateway
CLM	Communications Link Module
CM	Computing Module
EPLCG	Enhanced Programmable Logic Controller Gateway
ENIM	Enhanced Network Interface Module
GUS	Global User Station
HG	Hiway Gateway
HM	History Module
NG	Network Gateway
NIM	Network Interface Module
PCNM	Personal Computer Network Manager
PLCG	Programmable Logic Controller Gateway
PLNM	Plant Network Module
SAM	Scanner Application Module
US	Universal Station
UxS	Universal StationX

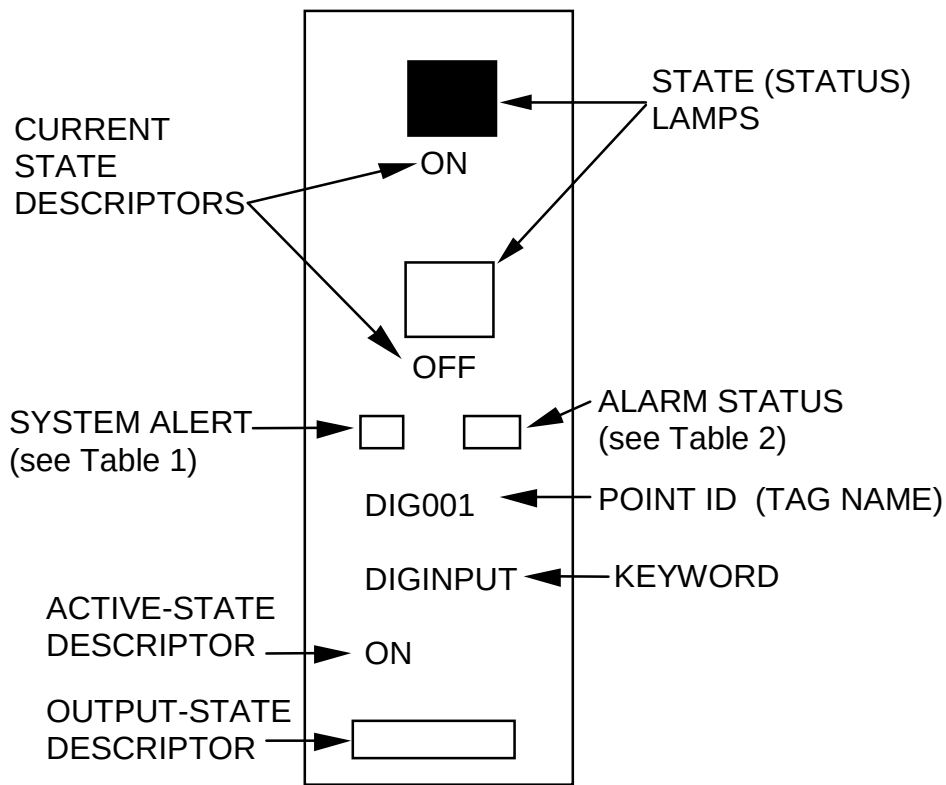
Hiway Devices

A-MC	Advanced Multifunction Controller
CB	Basic Controller
CPC	Critical Process Controller
DHP	Data Hiway Port
EC	Extended Controller
GCPI	General Purpose Computer Interface
HG	Hiway Gateway
HIM	Hiway Interface Module
HLPIU	High Level Process Interface Unit
HTD	Hiway Traffic Director
LEPIU	Low Energy Process Interface Unit
LLPIU	Low Level Process Interface Unit
MC	Multifunction Controller
PCSI	Personal Computer Serial Interface
PIU	Process Interface Unit

ALARM AND STATUS INDICATORS

Group and Detail Display Indicators





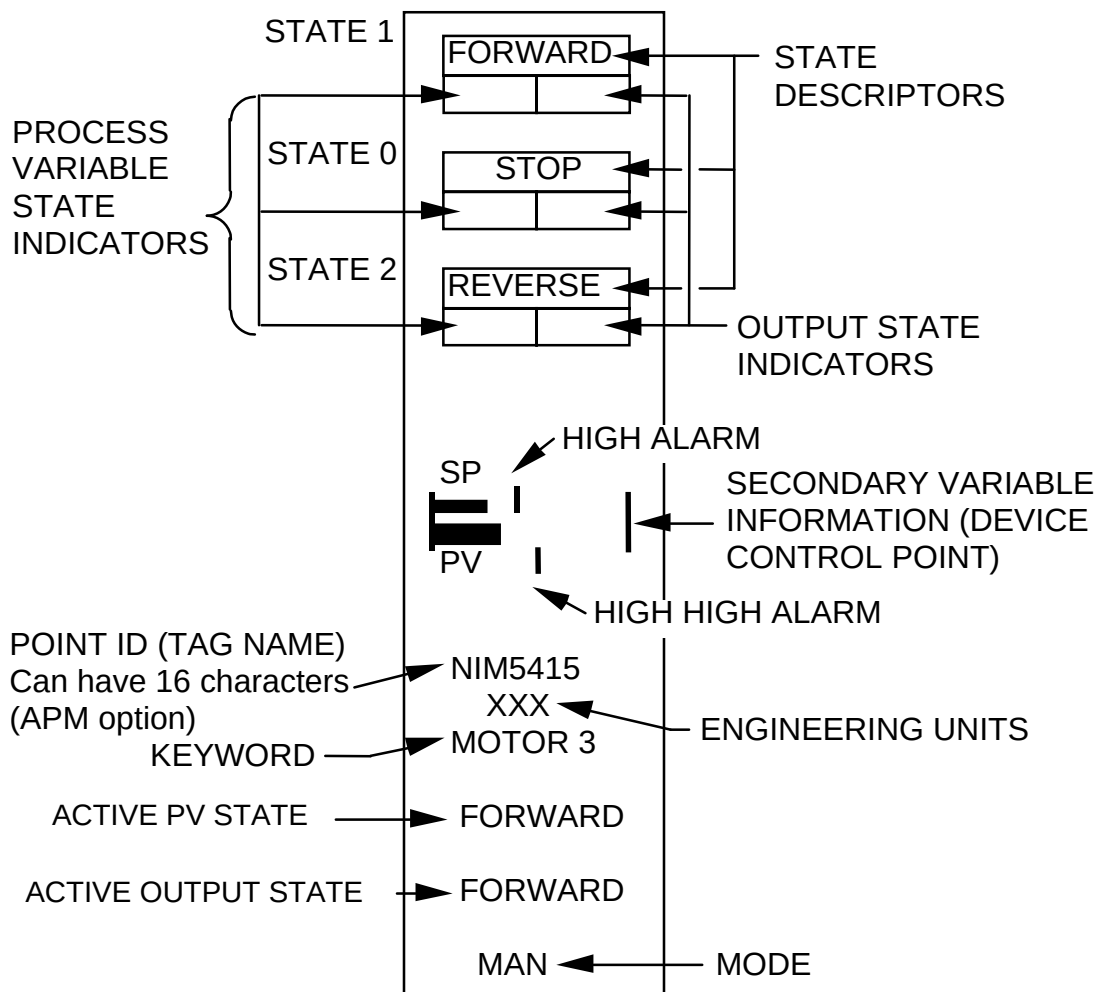
STATUS CODES/NOTES

- ????? = Values not accessible
- ----- = invalid state/input
- RESET = Hiway box in reset state
- CL SWITCH points show S1 or S2 Request Status for SP, and S1 or S2 Current Status for OP

NOTES

- Points with two inputs/two outputs show only inputs as state lamps
- Switch-type points can show up to four state lamps
- For dual-digital inputs: INBETWEEN = both state lamps filled FAULTY = both state lamps empty

DIGITAL/ FLAG/ CL SWITCH-POINT INFORMATION ON GROUP/ DETAIL DISPLAY



DIGITAL COMPOSITE AND DEVICE CONTROL POINT INFORMATION ON GROUP/DETAIL DISPLAY

TABLE 1 — SYSTEM ALERT INDICATORS

INDICATOR	MEANING
S	Point's box database saved (UAC Reserve Controller)
I	Point is in IDLE state; MC, XPM, LM
F	Point failed; PIU, MC, XPM, LM
U	DHP not scanning point; possible bad PC-configuration address
R	Point is in RESET state; Data Hiway points
NOTE: All indicators displayed in yellow	

TABLE 2 — ALARM STATUS INDICATORS

INDICATOR	MEANING	SOURCE DEVICES
[BLANK]	Point not in alarm or status not accessible	All devices
* (yellow)	Overview Tracer: indicates when point's PV deviation from the setpoint is greater than the overview value	AM, XPM
A	Flag Alarm State	HG, AM
ADV	Advisory Deviation	AM, XPM
BC	Bad Control Flag Alarm	AM, XPM
BGI	Bad General Input	AM
BP	Bad Process Variable	AM, XPM
BSV	Bad Secondary Variable	AM, XPM
C1, C2 C3, C4	Custom Alarms	
CLE	CL Error	AM, XPM
CLF	CL Fail	AM, XPM
CMF	Command Fail	XPM
CNF	Configuration Alarm	AM, XPM
COS	Digital Change of State	XPM
DFN	Digital Off Normal	AM, XPM
DIS	Disabled alarm enable status (R510 and later)	AM, NIM, HG
NOTE: Indicator color (red or yellow) corresponds to alarm's priority. For example: Red = Emergency, Yellow = High or Low		

TABLE 2 — ALARM STATUS INDICATORS (continued)

INDICATOR	MEANING	SOURCE DEVICES
0DV	Preset Alarm (Timer/Counter Points)	AM, EC
1DV	Pre-Preset Alarm (Timer/Counter Points)	AM, EC
2DV	Pre-Pre-Preset (Timer/Counter Points)	AM, EC
FB	Digital Feedback (Command Disagree)	XPM
HD	Deviation High	HG, AM, XPM
2HD	Deviation High High	HG
3HD	Deviation High High High	HG
HO	Output High	XPM
HP	PV High	HG, AM, XPM
2HP	PV High High	AM, EC, XPM
3HP	PV High High High	EC
HS	Secondary Variable High	APM, HPM
2HS	Secondary Variable High High	APM, HPM
INH	Alarms are inhibited (overwrites display of alarm status indicator)	All devices
INV	Invalid Alarm State	HG, AM, XPM
LD	Deviation Low	HG, AM, XPM
2LD	Deviation Low Low	HG
3LD	Deviation Low Low Low	HG
NOTE: Indicator color (red or yellow) corresponds to alarm's priority. For example: Red = Emergency, Yellow = High or Low		

TABLE 2 — ALARM STATUS INDICATORS (continued)

INDICATOR	MEANING	SOURCE DEVICES
LO	Output Low	XPM
LP	PV Low	HG, AM, XPM
2LP	PV Low Low	AM, EC, XPM
3LP	PV Low Low Low	EC
OFN	Off Normal	XPM
OH	Output High	XPM, HG
OL	Output Low	XPM, HG
ORC	Output Rate of Change	HG, PM, APM
OTC	Open Thermocouple	PIUs
OV0	Interlock Tripped, State 0	APM, HPM
OV1	Interlock Tripped, State 1	APM, HPM
OV2	Interlock Tripped, State 2	APM, HPM
PSC	PV Significant Change	AM, XPM
RC+	Positive Rate of Change	AM, XPM
RC-	Negative Rate of Change	AM, XPM
ROC	Rate of Change	EC
S4	S4 Logic Alarm	EC, AM, XPM
SHD	Computer Shed	HG, AM, XPM
SI0	Safety Interlock Tripped, State 0	APM, HPM
SO	Schedule Overrun	AM
SW1 - SW3	Switch Alarms	HG, AM, XPM
TMO	Timeout (Timer/Counter)	HG, AM, XPM
UNC	Uncommanded Change	XPM
UNR	Unreasonable PV	XPM
NOTE: Indicator color (red or yellow) corresponds to alarm's priority. For example: Red = Emergency, Yellow = High or Low		

TABLE 3 — SETPOINT STATUS INDICATORS

INDICATOR	MEANING
*	When point is in manual mode, setpoint is only used as target value. No control action occurs.
Λ	Setpoint cannot be increased due to limiting on output.
V	Setpoint cannot be decreased due to limiting on output.
B	Bad Setpoint Value
H	Setpoint High Limit
L	Setpoint Low Limit
P	Target value processing is Preset: AM only
R	Target value processing is Running: AM only

TABLE 4 — PV STATUS INDICATORS

INDICATOR	MEANING
Blank	Normal
Red B	Bad PV Value
Red H	Bad PV and PV High Extended Range Violation
Red L	Bad PV and PV Low Extended Range Violation
Red M	Bad PV and PVSOURCE is MANual
Red S	Bad PV and PVSOURCE is Substituted
Yellow U	PV is Uncertain
Yellow H	PV is Uncertain and PV Clamped to High Extended Range
Yellow L	PV is Uncertain and PV Clamped to Low Extended Range
Yellow M	PV is Uncertain and PVSOURCE is MANual
Yellow S	PV is Uncertain and PVSOURCE is Substituted

TABLE 5 — OUTPUT STATUS INDICATORS

INDICATOR	MEANING
*	Indicates that integral ramping of the controller has been stopped.
H	OP High Limit
L	OP Low Limit
Λ	Secondary of this point is at setpoint or output High limit. Output is only allowed to decrease.
V	Secondary of this point is at setpoint or output Low limit. Output is only allowed to increase.

TABLE 6 — POINT STATUS INDICATORS

INDICATOR	MEANING
BAD TYPE	Point type cannot be configured in Group.
BYPASS	Permissives and override interlock can be bypassed.
MISMATCH	Database does not match Area database.
INACTIVE	Point is Inactive.
OUT SERV	Point is Out Of Service.
RED TAG	Point is Red Tagged.
INIT	Point being Initialized, cascade broken, or output destination unavailable.
TEST	Point in test.
EXP	Timer Expired.
CNFG ERR	Point not present when database built.
SEL	Point Selected by Override Selector.
NOT SEL	Point Not Selected by Override Selector.
CFM	Confirmable Message on point (ACIDP point type).
xxxx	Program Abort Code for ACIDP in CM-60; xxxx (for example 2B17) indicates the specific reason for the point-processing abort. Refer to the CM60 Users Manual.

TABLE 7 — POINT STATUS INDICATORS

INDICATOR	MEANING
BAD TYPE	Point type cannot be configured in Group
BYPASS	BYPASS is ON
CFM	Confirmable Message on point (ACIDP point-type)
CNFG ERR	Point not present when database built
EXP	Timer Expired
INACTIVE	Point is Inactive
INIT	Point being initialized
MISMATCH	HG/AM database does not match Area database
NOT SEL	Point Not Selected by Override Selector
OUT SERV	Point is Out of Service
RED TAG	Point Red Tagged
SEL	Point Selected by Override Selector
TEST	Point in test
XXXX	Program Abort Code for ACIDP in a computing module; xxxx (for example 2B17) indicates the specific reason for the point-processing abort. Refer to the CM User Manual for further information.
[AAAAAAAA]	Up to 8-character status message configured by user
[Override Descriptor]	Configured override descriptor when override alarm is active
.....C	C following an override alarm indicates that confirmation is required

TABLE 8 — MODE INDICATORS

INDICATOR	MEANING
LOC MAN	Local (Loop) Manual; the Output is being set locally at the Hiway box
MAN	MANUAL: OP controlled by Operator
AUTO	AUTOMATIC: OP calculated by algorithm based on operator-controlled setpoint
CAS	CASCADE: SP or OP is from AM or CM or from point in same Hiway controller file ¹
BCAS	Backup CASCADE: SP or OP of Hiway controller point that was being set by an LCN Module (AM, CM), is now being set by a point in the same Hiway controller file ¹ ; in other words; the LCN module has shed control. BCAS applies to HG points only.
P MAN	Program MANUAL, OP controlled by CL/FORTRAN/ Pascal program in AM or CM
P AUTO	Program AUTO: SP controlled by CL/FORTRAN/ Pascal program in AM or CM
NOTES: 1- in the same controller; applies to CB and MC points. EC points can cascade to points in other boxes. C after the Mode means a CASCADE-request has been made; does not apply to AM. Mode in reverse-video (for example, MAN) means that the point IS NOT in its Normal Mode. Normal Mode is determined by configuration.	

continued

TABLE 8 — MODE INDICATORS (continued)

INDICATOR	MEANING
P CAS	Program CASCADE: SP or OP set by a CL program
P BCAS	Program BCASCADE: for secondary points in a cascade whose primary is in a different LCN Module; when the Module containing the Primary fails, the secondary point enters this Mode if CASCADE is the configured shed mode.
RUNNING STOPPED TIMEOUT COUNTOUT	Current operating state of XPM/AM- or DHP-based Timers, Counters, or Analog Accumulation points.
RUN RUN/REPT HOLD HOME JOG	Current operating state of a point with a CL Sequence
??????	Mode is inaccessible
!!!	Mode is indeterminate: PIU only
-----	Mode is invalid
MOD	Multifunction Controller's analog output is being controlled by one of the MCs Control Slots
NOTES: C after the Mode means a CASCade-request has been made; does not apply to AM. Mode in reverse-video (for example, MAN) means that the point IS NOT in its Normal Mode. Normal Mode is determined by configuration.	

System Status Display Indicators

TABLE 9 — NODE STATUS STATES

STATE	MEANING
OK	All nodes are either running, alive, qualified, or under test
FAIL	One or more nodes have failed
MAINT	One or more nodes have Maintenance Recommendations
ERRORS	One or more nodes have an Auxiliary Node Status

TABLE 10 — PROCESS NETWORK STATUS STATES

STATE	MEANING
OK	Boxes, devices, and cables are OK
FAIL	One or more boxes or Devices have failed
INIT	Hiway or UCN is starting up
????	Status is unavailable from Hiway or UCN
SWITCH	Cable switching is in progress
IDLE	One or more MCs are in IDLE or one or more XPMs or LMs are in IDLE

TABLE 11 — LCN CABLE STATUS INDICATOR

INDICATOR	MEANING
REVERSE-VIDEO	Active cable
GREEN	Normal condition (auto-cable swaps every minute)
YELLOW	Possible bad cable (auto-cable swaps every 10 min.)
RED	Total cable failure

LCN Node Display Indicators

TABLE 12 — LCN NODE STATUS INDICATOR

INDICATOR	MEANING
OK	Node is running "on-process."
BACKUP	Node is the backup for the other node of this node pair.
OFF	Node is not on the LCN, or power at node is off.
PWR_ON	Power is on but the node is not yet qualified or loaded.
QUALIF	Quality Logic Tests or self-tests are running.
READY	Node is qualified but not loaded.
FAIL	Power is applied but the node has failed.
TEST	Node has been placed in Test state to run its Maintenance Personality.

TABLE 13 — STATION TYPE INDICATOR

INDICATOR	MEANING
GUS	Global User Station with Universal Personality
OPR	Operator Personality
UNVL	Universal Personality
US	Universal Station power is on or off and personality is not loaded

Hiway Status Display Indicators

The following table lists the more common *informative* display indicators. *Failure* indicators (not listed in this table) are preceded by a double-dash; their detailed meaning can be determined by looking at the specific Box Status Display.

TABLE 14 — HIWAY BOX STATUS INDICATORS

MESSAGE	MEANING
OK	Box is functioning normally.
OK CP INHBT	Box is functioning normally, but automatic checkpointing has been inhibited for this box or hiway; manual checkpointing is still permitted.
*	Save or Load of checkpoint data is in progress.
ALM FAIL	Controller is no longer performing the Alarm Reporting function (EC & MC only).
BAD STATE	The box was not in a proper state for saving/loading data from/to the box.
FILE ERROR	An error occurred during loading of box data. This can be caused if the data is being loaded from write-protected removable media.
LOADED	Trend Program was fully transmitted over the hiway and the Trend Program is running.
NO MEMORY	Controller does not contain a memory card.
NO RESPONSE	Controller is not responding to continued hiway requests.
NOT ATTEMPT	Requested operation was not attempted because an error condition was detected.

Hiway and UCN Status Display Indicators

TABLE 14 — HIWAY BOX STATUS INDICATORS (continued)

MESSAGE	MEANING
NOT LOADED	Controller did not allow startup of the Trend Program, or the Trend Program failed on startup.
NOW RUNNING	Trend program is currently running.
RESTORED	Data was successfully restored to the box from a checkpoint file.
SAVED	Box data was successfully saved in a checkpoint file.
TND RUNNING	Trend program is currently running.
TRND LOADED	Trend Program was fully transmitted over the hiway and the Trend Program is running.
WRONG TYPE	Box type configured disagreed with the type of box on the hiway.

TABLE 15 — UCN DEVICE STATUS INDICATORS

STATE	MEANING
OK	Running normally, no failures.
BACKUP	Device is operating in backup capacity with no failures, and running normally.
IOIDL	One or more I/O Processors are in IDLE. No failures.
IDLE	XPM or LM is in IDLE. Some I/O Processors may be in IDLE, but XPMIDL overrides IOIDL. No failures.
PARTFAIL	Partial failures. The XPMM or LM has one or more soft failures.

TABLE 15 — UCN DEVICE STATUS INDICATORS (continued)

STATE	MEANING
PF_IOIDL	There are one or more partial failures, and one or more I/O Processors are in IDLE.
PF_IDLE	There are one or more partial failures and the UCN device is in IDLE.
OFFNET	UCN device is not responding on the UCN because 1) it does not exist, 2) it is powered off, or 3) it has a hardware failure that prohibits it from responding.
POWERON	UCN device is responding on the UCN, but has not yet run its self-tests. This is a transitory state that occurs when the XPMM is powered on before the self-tests finish.
ALIVE	UCN device is responding on the UCN, but is neither loaded nor functioning.
LOADING	UCN device is being loaded.
NOTCONFIG	There is no UCN device configured at this UCN node address.
CONFGMIS	Node type that exists at this UCN address does not match the type that is configured for this address.
NOSYNCH	Secondary UCN device is running, but has failed to synch its database with the primary.
FAIL	The UCN device has experienced a hard failure, and therefore access to I/O Processor status and data is unavailable. UCN device control functions have ceased; effectively a total UCN device failure.
COMMFAIL	NIM cannot communicate with any other UCN node.
UNKNOWN	When the primary NIM is in the COMMFAIL state, all other nodes on that UCN are in the UNKNOWN state until the primary NIM reestablishes communication.

XPM Status Display Indicators

TABLE 16 — XPM DEVICE STATUS INDICATORS

STATE	MEANING
OK	No failures, running normally.
BACKUP	Device is operating in backup capacity with no failures and running normally.
BKUP_SF	Backup XPM device has detected a soft failure.
IDLE	XPM device is idle, and contains no failures.
SOFTFAIL	XPM device contains one or more soft failures.
IDLESF	XPM device is idle and contains one or more soft failures.
OFFNET	XPM device is not responding on the UCN because 1) it does not exist, 2) it is powered off, or 3) it has a hardware failure that prohibits it from responding.
POWERON	XPM device is responding on the UCN, but has not yet run its self-tests. This is a transitory state that occurs when the device is powered on before the self-tests finish.
ALIVE	XPM device is responding on the UCN, but is neither loaded nor functioning.
ALIVESF	XPM device is ALIVE and contains one or more soft failures
LOADING	XPM device is being loaded.
NOTCONFIG	There is no XPM device configured at this UCN node address.
CONFGMIS	The XPM device type that exists at this UCN address does not match the type that is configured for this address.
NOSYNCH	Backup XPM device is running, but has failed to synch its database with the primary.
FAIL	XPM device has experienced a hard-failure and is totally nonfunctional.
UNKNOWN	Transitional status: the XPM device is changing from one state to another.
ATTENTION	With TPN R684, when the EHPM or ENIM is configured and loaded, the text EHPM or ENIM is displayed respectively in the auxiliary status area on the UCN Status Display. This auxiliary status area of the UCN node shows any one of the following • Control State status – when the Control States target is selected. • Auto Checkpoint state – when the Auto Checkpt target is selected. • UCN Cable status - when the UCN Cable Staus target is selected • EUCN Node Type – when no target is selected or when the Load/Save Restore or Run States target is selected

TABLE 17 — I/O PROCESSOR STATUS INDICATORS

STATE	MEANING
OK	No failures, running normally.
IDL	IDLE. Node is idle, and contains no failures.
SF	SOFTFAIL. Node contains one or more soft failures.
IDLESF	Node is idle and contains one or more soft failures.
POWERON	Processor is on the I/O Link, but self-tests are not completed. This is a transitory state that occurs when power is turned on.
NR	NORESP. Processor firmware does not respond to the I/O Link communications, but hardware acknowledges messages. Indicates a hard processor failure.
NONEXIST	Processor firmware does not respond on the I/O Link.
COMMERR	Cannot communicate with the processor (communications error)
CFG	CONFGMIS. The I/O Processor type that exists in this card slot does not match the type that is configured for this slot.
IOP COMM	Serial Interface input/output processor is not present or XPM is in process of loading array point configuration to serial interface input/output processor (transient state).
BKP	Redundant IOP card is backing up the primary IOP card.
DB_VALID	Database is valid.
DBINVALD	Database is invalid. Database has to be reloaded or restored into IOP.

ERROR MESSAGES

TABLE 18 — ERROR MESSAGES

MESSAGE	MEANING/CORRECTIVE ACTION
ACCESS LEVEL ERROR	The requested action cannot be performed unless the Universal Station keyswitch is switched to a higher level of access (SUPVR or ENG).
ILLEGAL BAD VALUE VALUE INVALID	An invalid entry was typed.
DATA ACCESS ERROR	Parameter does not exist for the selected point.
DATA NOT AVAILABLE	Data does not exist for the given point and time (history).
DEVICE RESET	The module or device where the point resides is in the RESET mode; no entry is allowed.
DISPLAY NOT CONFIGURED	The display requested does not exist on the system.
ENTITY NOT FOUND CANNOT FIND POINT	Tag name entered does not exist.
FAIL: USnn	US failed during a load. nn is node number of US.
INITIALIZATION	Selected parameter cannot be changed with point in current initialization mode.
INVALID ENTRY	An invalid entry was typed.
INVALID MODE PVSOURCE INVALID	Selected parameter cannot be changed with point in current mode.

TABLE 18 — ERROR MESSAGES (continued)

ATTRIBUTE	MEANING
INIT ERROR	The selected parameter cannot be changed because the point is in initialization mode.
LMT OR RANGE ERR	An attempt was made to change the value of a clamped parameter beyond the range limit.
NO CONFIGURED TREND	No trend data is available for the selected point.
PNT STATUS ERR	The module or device where the point resides has failed; no entry is allowed.
PRINTER NOT CONFIGURED	The print request could not be honored because the requested printer number is not valid.
PRIOR FUNCTION NOT COMPLETED	A previous function or activity must complete before the requested function can be started.
READ ONLY PAR	The selected parameter is for viewing only and cannot be changed by the operator.
RED TAG	All output entries are disallowed; the point is "Red-Tagged."
START TIME IS AFTER STOP START DATE IS AFTER STOP DATE OR TIME EXPECTED	The requested time base is incorrect.
SYS: UNABLE TO OPEN NCF FILE	NCF file is not on the network.
UNIT NOT CONFIGURED	Entered Unit ID does not exist or is not assigned to this station's Area.

POINT ATTRIBUTE DESCRIPTIONS

TABLE 19 — POINT ATTRIBUTE DESCRIPTIONS

ATTRIBUTE	MEANING
ALARM INHIBIT	Alarms are set to the INHIBIT state (no annunciation, no history collection, no journaling, and not shown on alarm displays).
ALARM DISABLE	Alarms are set to the DISABLE state (no annunciation and are not shown on alarm displays).
INACTIVE	Point execution status is set to INACTIVE.
MANUAL PV	PV source is set to MANUAL.
SUBSTITUTED PV	PV source is set to SUBSTITUTE (from external computation).
UNCERTAIN PV	Universal Station is receiving a questionable PV value.
BAD PV	Universal Station is receiving an invalid PV value.
TRACK PV	Currently operating in PV Tracking mode.
MANUAL MODE	Control mode is currently MANUAL.
BCAS MODE	Control mode is currently Backup Cascade (the assigned local data point has taken over as primary).
BCAS MODE	Control mode is currently Backup Cascade (the assigned local data point has taken over as primary).
CASCADE REQUEST MODE	A request for the point to enter cascade mode had been made, but a failure condition prevents implementation of the request.
PROGRAM MODE	Currently operating in Program mode; points in controller are operating under control of program.

TABLE 19 — POINT ATTRIBUTE DESCRIPTIONS (continued)

MESSAGE	MEANING
COMPUTER SHED	The high-level processor has been disengaged from the control loop.
INITIALIZED	Output destination unavailable.
BYPASS	Digital composite or device control point with an interlock or permissive that has been bypassed.
RED TAGGED	Tagged with a red marker; output change is disallowed; point in manual mode.
OVERRIDE ALARM W/CF	Override alarm that is awaiting confirmation.
The following attributes apply to XPM Process Module Points:	
LOADED	CL program is Loaded in Process Module.
NOT LOADED	CL program is Not Loaded in Process Module.
ERROR	Error has occurred during program execution.
FAIL	Failure has occurred during program execution.
HOLD	Operating status is HOLD.
SHUTDOWN	Operating status is SHUTDOWN.
EMERGENCY SHUTDOWN	Operating status is EMERGENCY SHUTDOWN.

INITIAL LOAD OF US FROM REMOVABLE MEDIA

Load with no US running and with no HM. US personality and area database available. This procedure is for an initial, one-time load of the US. For subsequent loads, use other load procedures.

Initial load from removable media

Action: Check for the prompt ">" at the US. If not present,

RESET



press under the hinged keyboard cover and wait for the ">" prompt to appear.

Action: Press:

LOAD

Response: **Prompt:** "N,1,2,3,4,X?"

Action: Select:

#

+

ENTER

Number of Drive

Response: Information will be loaded from removable media

Prompt: "OPR, UNP, X?"

Action: Type in OPR, or UNP **+**

ENTER

Response: Selected personality is loaded into your station

Prompt: "NCF? N,1,2,3,4,X?"

Action: Select:

#

+

ENTER

Number of Drive

Response: NCF is loaded into your station

Prompt: "ARnn N,1,2,3,4,X?"

Action: Select:

#

+

ENTER

Number of Drive

Response: Area database is loaded into your station

Prompt: "ABST N,1,2,3,4,X?"

Action: Select:

#

+

ENTER

Number of Drive

Response: Abstracts are loaded into your station

Prompt: "BUTT N,1,2,3,4,X?"

Action: Select:

#

+

ENTER

Number of Drive

Response: Button database is loaded. Station is now loaded

LOAD US FROM THE HISTORY MODULE

This procedure loads a US from an HM.

Loading from the Network

Action: Check for the prompt ">" at the US. If not present,
RESET  press under the hinged keyboard cover and wait for the ">" prompt to appear.

Action: Press: 

Response: **Prompt:** "N,1,2,3,4,X?"

Action: Press the <ALPHA SHIFT> key.

Response: the key's red light appears.

Action: Press:  + 

Response: Information will be loaded from Network (History Module)

Prompt: "OPR, UNP, X?"

Action: Type in OPR or UNP + 

Response: Selected personality is loaded from the Network into your station.

Response: **Prompt:** "NCF? N,1,2,3,4,X?"

Action: **Press:**  + 

Response: NCF will be loaded from Network (History Module) into your station.

Response: System Status display appears when loading is nearly completed.

LOAD A GUS

This procedure loads a GUS.

Loading from the
Network

Action: Check for the prompt ">" at the GUS.

Action: Select **LOAD** target from Native Window

Response: **Prompt:** "W, N,1,2,3,4,X?"

Action: Press: **W** + **ENTER**

Response: **Prompt:** "NCF? N,1,2,3,4,X?"

Action: **Press:** **N** + **ENTER**

Response: NCF will be loaded from Network (History Module)
into your GUS.

Response: System Status display appears when loading is
nearly completed.

LOAD A US FROM ANOTHER US IN THE CONSOLE

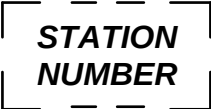
Load Operator or Universal Personality

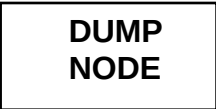
There are two methods of loading a US from another US on the same console.

Note: *The US must be in the FAIL, PWR_ON, or QUALIF state.*

METHOD A: This method allows the user to choose loading from network or removable media, manually or automatically.

1.  OR  at a US,
or select a US node on the System Status
display then select the [NODE STATUS]
target.

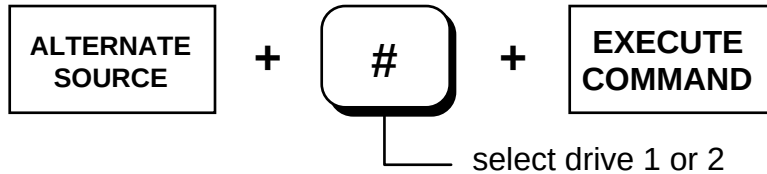
The Console Status and Assignment display
appears.
 2.  select from the display the US to be loaded.

 Then select the Load/Dump target.
Four hidden targets appear.
 3.     select
load type
(But not
Dump
Node)
- AUTOLOAD NET**—Select and press ENTER. Loads the target US from the Network. (No other steps.)
- AUTOLOAD LOCAL**—Select and press ENTER. Loads the target US from removable media or Network. (See step 5.)
- MANUAL LOAD**—Select and press ENTER. Select the personality type, and load from Network or removable media. (See step 4.)
4.  OR  select a Personality for loading

5. Select program load source:



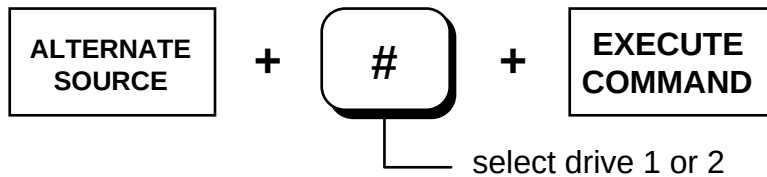
Insert the removable media with Operator or Universal Personality into drive 1 or 2 and select:



6. Select data load source:



Insert the removable media containing with your NCF (&ASY) into drive 1 or 2 and select:



7. **ENTER** to begin loading.

The System Status Display appears when loading is near completion.

METHOD B: This method automatically loads the US from another US that is in the same console.

1.  **at a US.**
2. Select the US node to be loaded.
3.  +  The US loads from the network.

The US status changes to “OK” on the System Status Display and the Console Status and Assignment Display.

LOAD US FROM A FAST LOAD REMOVABLE MEDIA

This procedure is intended for use in the event of a complete TPS system shutdown. A specially prepared removable media is required—see the Engineer's Reference Manual (subsection-Node Loading and Node Dumps.)

Fast Load from removable media:

1.  Press the **LOAD** button at US/GUS. The "**N,1,2,3,4,X?**" prompt appears.
2. Insert a Fast Load removable media (Section 20 of the Engineer's Reference Manual) in either drive 1 or 2.
3. Select the drive you are using by entering either "**1**" or "**2**", then press



Personality load begins.

If there is only one personality on the removable media, the personality will automatically load.

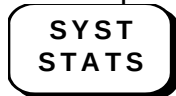
If there is more than one personality on the removable media, go to the next step.

4. "OPR, UNP?" If both personalities are on the removable media, this prompt appears. Key in your choice of either "**U**" for Universal Personality or "**O**" for Operator Personality. Press ENTER.

The System Status Display appears. Loading of the US is complete. If you need to use this removable media to reestablish a "path to the valve," continue with the next step.

- 5 On the Operator's Keyboard select

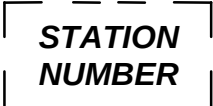
The System Status display appears.



-
- | | | |
|-----------|--|--|
| 6. | On the System Status display, select a NIM or HG node. | The node box is backlit. |
| 7. | Select the [LOAD SELECT] target and press [ENTER]. | The NIM or HG and the nodes on the associated process network are loaded. When loading is complete, the NIM or HG status changes to OK. You can now resume control of the points on the process network. |
-

LOAD A US FROM REMOVABLE MEDIA USING THE SCREEN TARGETS

This procedure loads a US from removable media using the screen targets.
Note: The US must be in the FAIL, PWR_ON, or QUALIF state.

Load a US from removable media	Action: Select:  Response: The Console Status display is called up Action: Select:  Response: Station to be loaded is selected Action: Select:  Response: The following target appears Action: Select:   Response: The following targets appear Action: Select:  Response: Drive targets appear Action: Select: Drive number 1, 2, 3, etc. +  Response: The following target appears Action: Select:  Response: Drive targets appear Action: Select: Drive number 1, 2, 3, etc. +  Response: Prompt: PRESS ENTER TO EXECUTE Action: Select:  Response: Station begins loading
---------------------------------------	--

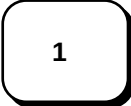
LOAD A US FROM REMOVABLE MEDIA USING THE KEYBOARD

This procedure loads a US from removable media using the keyboard

1. **RESET**  Press the **RESET** button at the Universal Station and wait for the ">" prompt to appear.

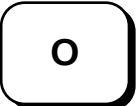
2.  Press the **LOAD** button at the Station.
The "**N,1,2,3,4,X?**" prompt appears.

Insert the removable media into drive 1 and select:

3.  + 

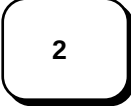
Prompt: "**UNP,OPR?**"

This prompt only appears if both personalities are on the removable media otherwise, the one residing personality is loaded automatically.

4.  +  OR  + 

The "**NCF? N,1,2,3,4,X?**" prompt appears.

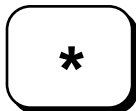
5. Insert the removable media with your NCF (&ASY) into drive 2.

6.  + 

System Status display appears.

CONTINUOUS OPERATING DISPLAYS

General Operating Keys



User-configured function key is configured for a designated action



Calls up an Associated Display



Calls up a preconfigured help display associated with the current display



Calls up the display that was shown immediately before the current display



Calls up the next higher numbered display, for example, going from group 1 to group 2



Calls up the next lower numbered display



Clears an incorrect or unwanted operator entry at any time before the ENTER key is pressed



Enters any information that has been typed



These four keys are used to position the cursor vertically or horizontally to any position on the display



Selects an item at the current cursor position on a display



Enables fast screen update on a group or detail display

Schematic

Call up procedure	Action: Press:  Response: Prompt: ENTER SCHEMATIC NAME Action:  +  └ type in the schematic name Response: Schematic display appears on the screen
Target selection for a schematic	Action: Use touch screen or Press:  to select screen targets    +  Response: Selected target action is initiated

Group

Call up procedure	Action: Press:  Response: Prompt: ENTER GROUP NUMBER Action:  +  └ type in group number 1-400 Response: Group display appears on the screen
Selecting a point on a group display	Action: Use touch screen or Press:  to select screen targets    +  Response: Selected point is highlighted

Detail

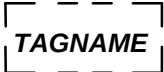
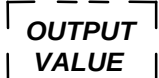
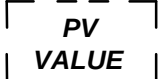
Call up procedure from a group display	Action: Press: TAGNAME + DETAIL Response: Detail display appears on the screen
Call up procedure from any display	Action: Select: DETAIL Response: Prompt: ENTER TAGNAME Action: a # + ENTER type in tag name
Page through a detail display	Complete the step above to call up a detail display Action: Select: PAGE FWD Response: To go to the next page on the detail display Action: Select: PAGE BACK Response: To return to the previous pages on the detail display
Navigate from detail display to other displays	Action: Select: FIRST PAGE on Detail Display Response: The following targets may appear: Action: Select: FIRST CTL ALGO CTLCONCT CONFIG CL BLOCK CUSTOM PRIOR GROUP GRPASSOC GRPHelp ASSOC CDETAIL Response: Navigates to other operator displays

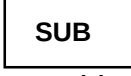
See Point Parameter Changes (following pages) for details on **selecting and changing parameters**.

With Experion R410, you can view and control the all the parameters, using Experion TPN point detail displays as you can with the Native Window point detail displays. While the Native Window display provides the data in a series of separate pages, Experion station display provides the same data as a set of tabbed displays. The tabbed displays framework of Experion Station window incorporates all the components that are supported in the Native Window.

POINT PARAMETER CHANGES

First select the point at a Group Display, or call up the Detail Display. The following are procedures to change various parameters by pushing buttons or using touch targets.

Changing Modes	Action: Select:  Response: Point becomes highlighted Action: Press:  OR  OR  Response: Operating mode changes to MAN, AUTO, or NORM
Changing Analog Setpoints	Action: Select:  Response: A text box appears above the setpoint Action: Type in new value +  OR Select one of the following buttons     Response: Setpoint changes only if point is in its proper mode
Changing Analog Outputs	Action: Select:  Response: Point becomes highlighted Action: Type in new value +  OR Select one of the following buttons     Response: Output value changes only if point is in its proper mode
Changing Analog Process Variables 	Action: Select:  Response: A text box appears above the PV Action: Type in new value +  Response: Process Variable value changes <i>(Can be entered only if PV Source and point are in Manual mode.)</i>

Changing the PV Source mode (Point must be in MANual mode) 	Action: Select:  Response: The following targets appear Action: Select:  OR  +   OR  Response: Process Variable Source mode changes
Changing Digital States: Data Hiway and UCN Points 	Action: Select:  Response: Digital Point becomes highlighted Action: Press:  OR  Response: New state appears above OP% field Action: Press:  Response: Output state changes
Changing Tuning Constant from a detail display 	Action: Select:  Response: Text box appears Action: Type in new value +  Response: New tuning constant is entered
Activate or deactivate an AM, XPM, or CM point from a detail display	Action: Select:  Response: The following targets appear Action: Select:  OR  Response: Selection becomes highlighted Action: Select:  Response: Point becomes active or inactive
Changing all other detail display parameters  OR 	Action: Select:  OR  Response: Text box appears Action: Type in new value +  Response: New values are now entered

ALARM FUNCTIONS

Silence the Horn

Silence the horn from any display	Action: Select:  Response: Alarm horn stops NOTE: With R650, in View Only mode, only the horn in the local station stops. In Process Operator mode (OPER), you can silence horns console-wide or area-wide. The action of the silence button is determined during NCF configuration.
-----------------------------------	--

Acknowledge Local Console Alarms

Acknowledge all US alarms	Action: Select:  Response: Console Status display appears on the screen Action: Select:  Response: Acknowledges all US alarms
---------------------------	--

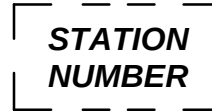
Acknowledge
PERIPHS OR
MAINT alarms

Action: Select:



Response: Console Status display appears on the screen

Action: Select:



Response: Station number becomes highlighted

Action: Select:



OR



Response: The corresponding display appears on the screen

Action: Select:



Response: Peripheral Status or Maintenance Information alarms become acknowledged and stop blinking

Acknowledge System Alarms

Acknowledge all System Status alarms	Action: Select:  Response: System Status display appears on the screen Action: Select:  Response: System Status alarms become acknowledged
--------------------------------------	--

Acknowledge Process Network (Hiway/UCN) Alarms

Acknowledge all Process Network alarms	Action: Select:  Response: System Status display appears on the screen Action: Select one of the NIM or HG node object boxes. Response: The node object box is backlit. Action: Select the  target. Response: The Process Network's Node Status display appears on the screen Action: Select:  Response: Hiway/UCN and Process-Connected Device alarms become acknowledged
Call up detailed information for an XPM and Hiway devices	Action: Select:  for Hiway devices Response: Text box appears Action: Type in box address 1-63 +  Response: Detail page of the box appears on the screen Action: Select:  +  Response: Detail page of the XPM appears on the screen

Lost Event (Alarm) Recovery

At the Alarm Annunciator Display, Area or Unit Alarm Summary Displays, or from System Status Display with a blinking EVENT RECOVERY target:

Call up lost events	Action: Select:	EVENT RECOVERY
	Response:	All lost events are notified
	Action: Select:	ENTER
	Response:	All lost events are recovered

System Alarm Disable

System alarms can be disabled/enabled at the node/network/ IOP level. The system alarms can be disabled/enabled through one of the following methods:

Call up System Status display	METHOD: 1	
	Action: Select	SYST STATS target.
	Response:	Standard System Status display appears.
	Action: Select	DISABL SYS ALM
OR		
Call up procedure	METHOD: 2	
	Action: Press	SCHEM on the keyboard.
	Response:	You are prompted to enter the schematic name.
	Action: Type \$ALLANP and press	ENTER
Response: DISABLE/ENABLE/VIEW ANNUNCIATION POLICY page is displayed		

You can disable/enable/view annunciation policy for LCN Nodes through one of the following methods:

From
DISABLE/ENABLE/VIEW
ANNUNCIATION
POLICY page

METHOD: 1

DISABLE/ENABLE/VIEW
ANNUNCIATION POLICY
FOR LCN NODES

Action: Select

Response: LCN Node Annunciation policy page
appears

Action: Select the target YES to disable the
System alarms for the required Nodes

Response: The YES target toggles to NO and is
displayed in Red

Action:

SELECT THIS TARGET TO PROPAGATE
ANNUNCIATION POLICY TO CONSOLE

Select

Response: The changes made to the local
annunciation policy are propagated to
all the US/GUS stations on the
console

OR

Call up procedure

METHOD: 2

SCHEM

Action: Press

on the keyboard

Response: You are prompted to enter the
schematic name

Action: Type \$ALLANP and press

ENTER

Response: LCN NODE ANNUNCIATION POLICY
page is displayed

Follow the steps mentioned in METHOD: 1

From DISABLE/ENABLE/VIEW ANNUNCIATION POLICY page	<i>METHOD: 1</i> Action: Select DISABLE/ENABLE/VIEW ANNUNCIATION POLICY FOR UCN/HIWAY Response: Process Network Annunciation policy page is displayed Action: Select the text box shown against PROCESS NTWK NBR Response: ENTER Process Network Number is displayed Action: Enter the Process Network Number Response: The nodes available in the network are displayed Action: Select the YES target to disable the System alarms for the required Nodes Response: The YES target toggles to NO and is displayed in Red Action: Select SELECT THIS TARGET TO PROPAGATE ANNUNCIATION POLICY TO CONSOLE Response: The changes made to the local annunciation policy are propagated to all the US/GUS stations on the console
OR	
Call up procedure	<i>METHOD: 2</i> Action: Press  on the keyboard Response: You are prompted to enter the schematic name Action: Type \$NWANP and press  Response: PROCESS NETWORK ANNUNCIATION POLICY page is displayed Follow the steps mentioned in METHOD: 1

You can disable/enable/view annunciation policy for IOP modules through one of the following methods:

From DISABLE/ENABLE/VIEW ANNUNCIATION POLICY page	METHOD: 1 DISABLE/ENABLE/VIEW ANNUNCIATION POLICY FOR IOP MODULES Action: Select Response: IOP Annunciation policy page appears Action: Enter the UCN address, Node address, and IOP address to be disabled and select ADD Response: The details are added to the list Action: Select SELECT THIS TARGET TO PROPAGATE ANNUNCIATION POLICY TO CONSOLE Response: The changes made to the local annunciation policy are propagated to all the US/GUS stations on the console
OR	

Call up System Status display

METHOD: 2

Action: Select  target.

Response: Standard System Status display appears

Action: Select the Network and select **NTWK/HWY STATUS** target

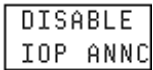
Response: UCN Status display appears

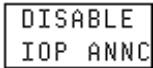
Action: Select a UCN node and select **DETAIL STATUS**

Response: xPM Status Display appears

Action: Select a node and select



Response:  target is enabled

Action: Select  target

Response: IOP Annunciation policy page appears with the currently disabled IOPs

Action: Enter the UCN address, Node address, and IOP address to be disabled and select **Add**

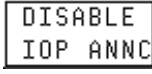
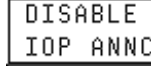
Response: The details are added to the list

Action: Select



Response: The changes made to the local annunciation policy are propagated to all the US/GUS stations on the console

OR

Call up procedure	<i>METHOD: 3</i> Action: Press  on the keyboard. Response: You are prompted to enter the schematic name Action: Type \$IOPANP and press  Response: IOP Annunciation policy page appears with the currently disabled IOPs Follow the last two steps mentioned in METHOD: 2
OR	
FROM xPM status display	<i>METHOD: 4</i> Action: Without selecting an IOP, select the  and  Response: IOP Annunciation policy page appears with the currently disabled IOPs Follow the last two steps mentioned in METHOD: 2 OR <i>METHOD: 5</i> Action: Select an IOP, followed by  and  Response: IOP Annunciation policy page is displayed with the selected UCN address, Node address and IOP address in the IOP TO DISABLE field Note: The IOP TO DISABLE field is filled with details only when the keylock is set to ENGR Follow the last two steps mentioned in METHOD: 2

System alarm disable - Error Messages in IOP Annunciation policy page

Error Messages	Meaning/Corrective Action
SLOT INVALID	ADD target was selected when there was no empty row in the IOP Annunciation policy page.
INTEGER TYPE EXPECTED	Alphabetic characters, special characters, or extra characters were entered in the Network/Node/IOP address port.
DUPLICATE	The selected IOP was already disabled.
STORE NOT VALID	An attempt was made to store the value OFF to the \$IOPADD, IOPDEL(nn), and \$ANPPROP parameters.
NUMBER INVALID	Zero was entered in the Network address/Node address/IOP address port.

Acknowledge Process Alarms

Acknowledge from the Area Alarm Summary display	Action: Select:  Response: Alarm Summary display appears on the screen Action: Select:  Response: All Process Alarms on that display become acknowledged
Acknowledge from the Alarm Annunciator display	Action: Select:  Response: Alarm Annunciator display appears on the screen Action: Select:  Response: All emergency priority alarms on that display become acknowledged
Acknowledge from the Unit Alarm summary	Action: Select:  Response: Text box appears on the left side of the screen Action: Type in unit ID Response: Page one of the Unit Alarm summary display appears on the screen Action: Select:  Response: All Unit alarms for the display page become acknowledged

Call up Unit Alarms from any process alarm display	Action: Select:  Response: Unit Alarm Summary is called up
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Acknowledge Messages

Acknowledge Messages	Action: Select:  Response: Message Summary display appears on the Screen Action: Select:  Response: All Messages on that page become acknowledged. Use the page forward and page backward keys to go through all pages on the Message Summary display
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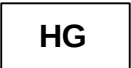
Confirm and Clear Messages

Confirm Messages	Action: Select:  Response: Message Summary display appears on the Screen Action: Select:  to confirm the message (Message must be acknowledged before confirmation)  to clear all acknowledged and confirmed messages on the page Response: Depending on selection, messages become clear or confirmed
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TREND FUNCTIONS

Group Trend

At an operating Group Display

Call up a group trend	Action: Press:  to display preconfigured Group Trend point OR  +  to add points to the Trend Display Response: Trend display appears on the screen
Select and change time base	Action: Select:  to select a new time base for the Trend Display Response: The following scrolling targets appear (HM Trends only) Action: Select:  OR  Response: Time base changes
Change trend data source	Action: Select:   Response: The following targets appear to the left of the trend graph      <i>Note: HG target is not included for XPM Trends</i> Action: Select one of the following targets to change data source for a single trend trace      Response: The data source changes for a single trace Or Action: Select:  +  Response: Change data source for all trend traces

Change scale range values	Action: Select: 100% OR 0% Response: Prompt appears to enter new high or low scale value Action: Type in new scale value + ENTER Response: Scale value changes
Re-range PV/SP	Action: Select GROUP key in the Operator keyboard and type the Group number and press ENTER Response: The Group Display gets displayed Action: Select a Tag name and press TREND key Response: Group Trend Display appears Action: Select the “Parameter change target” of a point Response: The “Parameter Change Menu” is invoked Action: Turn the PV/SP Parameters ON to trend them Response: PV/SP Parameters will be added for trending Action: Deselect and select a point tag name Response: EU range boxes appear for the point, with the values of the trended parameters Action: Select HI RANGE OR LOW RANGE Response: The prompt “ENTER NEW PV/SP RANGE” appears (the toggle target might have to be selected twice depending on its toggle state) Type in new scale value for PV/SP and press ENTER Note: The High re-range target and the Low re-range targets are independent of each other. Toggling the target on the High re-ranging target does not affect the Toggle State of the Low re-ranging target, and vice-versa. For example: if you select the High re-range target, you are prompted for the values of PV/SP (high). Now, if you activate the target for Low re-ranging, you are again prompted for the values of PV/SP (low) and not for OP.

Re-range OP	Action: Select GROUP key in the Operator keyboard and type the Group number and press ENTER Response: The Group Display gets displayed Action: Select a Tag name and press TREND key Response: Group Trend Display appears Action: Select the "Parameter change target" of a point Response: The "Parameter Change Menu" is invoked Action: Turn the OP Parameter ON to enable target action for OP re-ranging Response: OP Parameter will be added for trending Action: Deselect and select a point tag name Response: EU range boxes appear for the point, with the values of the trended parameters Action: Select HI RANGE OR LOW RANGE Response: The prompt "ENTER NEW OP RANGE" (the toggle target might have to be selected twice depending on its toggle state) appears Type in new scale value for OP and press ENTER Note: If OP is not added for trending, re-range targets will not have toggling action. These targets will continue to work as the targets for PV/SP re-ranging, as it was prior to R630.
Re-scale a trace	Action: Select: TAGNAME Response: PV scale appears on left side of screen Action: Select: HI RANGE OR LOW RANGE Action: Type in new scale value + ENTER Response: Selected tag name trace is rescaled for greater or lesser

Add trace to trend graph	Action: Select: TAGNAME of a point not being trended. Press TREND to add trend trace line of selected point tag Response: Trace added to the trend graph
Delete trace from trend graph	Action: Select: TAGNAME of a point being trended Press TREND to delete trend trace line from selected point Response: Trace deleted from the trend graph

Use scroll time target	Action: Select: DATE/TIME at lower right corner of trend graph Response: Date port appears Action: Type in new date + ENTER Response: Time port appears Action: Type in new time + ENTER Response: Trend scrolls to requested time and date NOTE: By pressing ENTER key without entering value, current date and time are used
Use center-line trending	Action: Select: R-AXIS Then select C-LINE Response: Center-line trend appears, with date and time target, showing values on either side of designated trend Time NOTE: When requesting a recent time trend, center-line trending will not allow future time to appear; instead, current time appears at the trend's right axis, and the new center-line time is adjusted to half of the time base off current time. NOTE: C-LINE trending can be used with other trending features.
Use hairline trending	Action: Select: R-AXIS Then select H-LINE Response: Hairline trend appears allowing the user to see the numeric data value at the hairline cursor, without going to the System Menu to retrieve history reports for the points trended

Group Trend Print

At the Group Trend Display...

Print and cancel print	Action: Press:	
	Response:	Trend printing is started
	Action: Press:	
	Response:	Trend printing is canceled

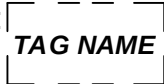
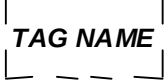
Standard Trend Pen Recording

At an operating Group Display...

Record and stop recording trends	Action: Press:	
	Response:	LED lights and trend starts recording
	Action: Press:	 again
	Response:	LED turns off and trend stops recording

Optional Enhanced Trend Pen Recording

If your system has the optional Enhanced Trend Pen Function:

Call up and change an enhanced trend pen record	Action: Press:  Response: Trend pen display appears and RECRD key LED lights
Add/change point/tag assignment	Action: Select:  or blank field on line where tag name will appear Response: Prompt ENTER POINT NAME Action: Type in tag name +  Response: Display updates to reflect change
Delete point/tag assignment	Action: Select:  Response: Tag name becomes highlighted Action: Type the word NULL +  Response: Point is now deleted
Change the trended parameter	Action: Select:  Response: Prompt ENTER PARAMETER NAME Action: Type in new parameter name +  Response: Trended parameter is now changed
Change the high or low limit	Action: Select:  OR  Response: Prompt ENTER NEW HIGH or LOW LIMIT Action: Type in new high or low limit +  Response: High or low limit is now changed

Change sampling rate	Action: Select: SAMPLING RATE Response: The following targets appear Action: Select: 4 SECONDS OR 1 SECOND OR Return to Normal Heading Response: Pen value sampling rate changes
Change pen status	Action: Select: PEN STATUS single pen OR Status all pens Response: The following targets appear Action: Select: Active Suspend Calibrate Return to Normal Heading Response: Display updates with new status
Change calibration % value	Action: Select: CALIBRATION % Response: Prompt ENTER CALIBRATION Action: Type in new calibration value (0 - 100)+ ENTER Response: Calibration value changes
Change the index	Action: Select: INDEX Response: Prompt ENTER INDEX Action: Type in new index value+ ENTER Response: Index value changes
Assign trend pen group	Action: Select: Group Response: Prompt ENTER TREND PEN GROUP NUMBER Action: Type in group number + ENTER Response: New trend pen group display appears

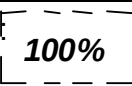
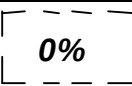
Hourly Averages

At an operating Group Display...

Call up hourly averages	Action: Press:  Response: The upper portion of the group display is replaced with hourly averages of up to 10 hours for each point
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ONE KEY CALL UP OF POINT TREND

Call up a Point Trend from Detail Display	Action: Press:  Response: Default trend display appears OR Select  from detail display Response: Menu appears Action: Select "MORE" target Response: Menu with TREND target appears Select  from the menu Response: Default Trend display appears
Call up a Point Trend from a standard display (i.e., Console Status display, System Status display, etc.)	Action: Press:  to display default trend display Response: Data port appears with "Enter point name" Action: Type in desired point name +  Response: Default Trend display appears
Call up a Point Trend from Node Point Summary with <u>Point selected</u> (i.e., Alarm Summary, Organizational Summary, etc.)	Action: Press:  to display default trend display Response: Default Trend display appears

Select and change Time Base	Action: Select:  Response: Time base menu and time base data port appears Action: Type desired number +  Response: New time base appears on trend display
Change Trend Data Source	Action: Select:  Response: All point parameter associated information is highlighted in the same color, and data source menu and data port appear Action: Type number of desired source (0,1 or 2) +  Response: Data source changes for trace appear
Change % Scale of Graph	Action: Select:  OR  Response: Prompt appears to enter new high/low scale value Action: Type in new scale value +  Response: Scale value changes and another prompt appears to enter new high/low scale value Action: Type in new scale value +  Response: Scale value changes

Change Engineering Units of Trace	Action: Select: <i>ENGINEERING UNITS</i> <i>high or low</i> Response: Prompt appears to enter new hi/lo range value Action: Type the new EU value + ENTER Response: New EU value is entered and prompt appears to enter new hi/lo range value Action: Type the new EU value + ENTER Response: Engineering Unit for point trace is rescaled for greater or lesser detail
Delete Trace	Action: Select: <i>POINT</i> <i>PARAMETER</i> Response: Delete Trace? prompt appears Action: Type Y + ENTER to delete Or Type N to cancel delete request Response: If delete was chosen, all remaining traces move one space to the left

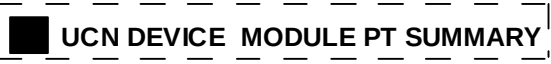
Add Trace	Action: Select: ADD TRACE Response: Prompt "Enter Point Name" with data port appears Action: Type point name + ENTER Response: Prompt "Enter Parameter Name" with data port appears Action: Type any parameter which can be trended + ENTER Response: New point parameter appears in the next available space to the right
Hairline Cursor Movement	Action: Select: HAIRLINE CURSOR MOVEMENT FACTOR to bring up factor menu. Response: Data port "Enter Factor Value (1-480)" and example factor menu appears Action: Type number for factor desired (1-480) + ENTER Response: The chosen factor appears Action: Select: Hairline Scroll Back Or Hairline Scroll Fwrđ Response: Hairline will move back or foward according to factor chosen

Trend Mode Change	Action: Select:  to change trend mode Response: Trend mode selection menu and "Enter Mode" data port. appear. Action: Type number corresponding to trend mode (R-Axis, C-Line, H-Line) desired  Response: New trend mode appears
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SYSTEM MENU FUNCTIONS

Organizational Summaries

Call up system menu	Action: Press:  Response: The system menu appears on the screen.
Manipulate targets in the organizational summary menu	Action: Select:  ORGANIZATIONAL SUMMARY MENU Response: Organizational summary display is called up.
	Action: Select:  UNIT POINT SUMMARY Response: Prompt: ENTER UNIT ID. Action: Type in unit ID +  Response: Unit point summary is called up.
	OR Action: Select:  HIWAY BOX POINT SUMMARY Response: Prompt: ENTER HIWAY NUMBER. Action: Type in Hiway number Type in Box number +  Response: Hiway Box point summary is called up.
	OR Action: Select:  POINT USAGE LIST Response: Prompt: ENTER POINT ID. Action: Type in tag name. Response: Point usage list is called up.

Manipulate targets in the organizational summary menu	OR Action: Select:  Response: Prompt: ENTER UNIT ID. Action: Type in unit ID +  Response: Unit Point Attribute Summary appears. OR Action: Select:  +  Response: Area Point Attribute Summary appears.
	OR Action: Select:  Response: Prompt TYPE IN NODE ADDRESS. Action: Type in node address +  Action: Select:  OR  Response: LCN node point summary is called up. OR Action: Select:  Response: Prompt ENTER UCN/DEVICE NUMBER Action: Type in UCN number ————— Action: Type in UCN device number ————— +  Action: Select:  OR  Response: UCN device point summary is called up. OR Action: Select:  Response: ENTER UCN/DEVICE/MODULE NUMBER. Action: Type in UCN number ————— Action: Type in UCN device number ————— +  Action: Type in Module number ————— Action: Select:  OR  Response: UCN device module point summary is called up.

Display or print menu items after each target selection	Action: Select: DISPLAY Response: Any of the targets selected will be displayed. Action: Select: PRINT Print target is optional. Action: Type in printer number + ENTER
Call up additional menu items	Action: When selected, the following target displays will appear. ☐ AREA TITLES ☐ UNIT TREND TITLES ☐ UNIT TITLES ☐ SCHEMATIC/FFL TITLES ☐ GROUP TITLES

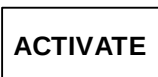
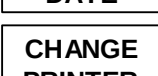
Reports, Logs, Journals or Printed Trends

Change reports, logs, journals, and printed trends

Action: Select:  **REPORT/LOG/TREND/JOURNAL MENU**

Response: The following targets appear.

Action: Select:

	to select a Report, Log, Journal, or Printed Trend
	to print the selected item
	to cancel the print operation
	to suspend periodic printing of the selected item
	to activate periodic printing of the selected item
	to display the selected item on the screen
	to select a new start time
	to select a new start date
	type in the printer number
	to change the printer assigned to the item

Response: Depending on target selection, changes will be made.

Call up additional menu items

 OR  to view all 132 columns of the display

Real-Time Journal Assignments

Call up additional menu items	Action: Select:  REAL TIME JOURNAL ASSIGNMENTS Response: The following targets appear. Action: Select:  to select the Real-Time Journal type  to suspend printing of the selected journal  to activate printing of the selected journal  type in the printer number   to change the printer assigned to the item Response: Depending on target selection, changes will be made.
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The following table lists the event types that can appear in the Real Time Journal:

TABLE 1 — Event Types

EVENT TYPE	DESCRIPTION
0	Fatal Driver Error or Crash Error
1	Non-fatal Driver Error or Software Error
2	Communication Error
3	CPU Driver Error
4	CRB Driver Error
5	FDS Driver Error
6	WDS Driver Error
7	PIC Driver Error
8	DHI Driver Error

TABLE 1 — Event Types (continued)

EVENT TYPE	DESCRIPTION
9	CLH Driver Error
10	SIO Driver Error
11	Floating Point Error
12	Exception Error
13	CLK Error
14	SPC Error
15	BL Error
16	PNI Error
17	NGI Error
18	WSI Error
19	Peripheral Error Event
20	Peripheral Event Recovery
21	Process Alarm
22	Sequence Event
23	CL Message
24	Process Alarm Change Event
25	CL Message Confirm
26	Values
27	SOE
28	Point/Unit/Console Alarm Disable/Inhibit, Unit Assignment, Sort, Filter, Suppress Audible, Freeze

TABLE 1 — Event Types (continued)

EVENT TYPE	DESCRIPTION
29	Operator Comment
30	Sequence Ack
31	Alarm Ack
32	CL Msg Acknowledge
33	SOE Ack
34	CL Msg Clear
35	Silence Alarm
36	LCN Event
37	Journal Box Event
38	Display Box Event
39	System Event Change
40	Off-process System Change Event
41	Operator-initiated LCN Change
42	Operator-initiated LCN Change
43	Operator-initiated Hiway Action Event
44	Operator-initiated Box Change
45	Operator-initiated Time Change
46	State Change Notification (Physical Node State, Logical Node State, Personality Change)
47	Slot Event
48	&ASY Restore Event

TABLE 1 — Event Types (continued)

EVENT TYPE	DESCRIPTION
49	Maintenance Recommendation
50	Corrective Action
51	Maintenance Action
52	Maintenance Clear
53	TAC Connect
54	TAC Disconnect
55	Maintenance Acknowledge
56	LCN Event Acknowledge
57	Hiway Acknowledge
58	Box Acknowledge
59	Peripheral Acknowledge
60	Cable Status
61	Lost Process
62	Lost Peripheral Event
63	Lost Hiway Event
64	Cable Status Acknowledge
65	Lost Process Event Acknowledge
66	Process Event Recovery
67	Lost Peripheral Acknowledge
68	Journal Overflow Operator Message
69	OPT (On Process Test) Error

TABLE 1 — Event Types (continued)

71	Lost Hiway Acknowledge
72	Hiway Event Recovery
73	Current Hiway Status
74	IOP Event
75	Current HPN Status Event
76	Auxilliary Status Event
77	Auxilliary Status Acknowledge
78	Operator HPN UCN Change Event
79	NIM, PM, APM, HPM, LM, IOP, Cable Acknowledge
80	HM Error Handler
81	HM Checkpoint Control
82	HM Operator Message System Event
83	Physical Node Status Change
84	Engineer Changes

Event History Menu

Set TIME SPAN, START DATE / START TIME, AND END DATE / END TIME Values in Event History (R600)

Action: Select  **EVENT HISTORY MENU**

Event History Retrievals display appears.

Action: Select START DATE/ START TIME value field

Response: TIME SPAN pop-up port appears.

Action: Enter number of hours for time span into pop-up port,
+ **ENTER**

Response: New time span value appears under TIME SPAN descriptor. END DATE / END TIME value field is set to current date and time. Start time in START DATE /START TIME value field is changed to a value which is end-time value minus time span value.

Change End Date and End Time to Current Date and Time	Action: Select END DATE / END TIME value field + ENTER Response: END DATE / END TIME value field is updated to current date and time. START DATE / START TIME value field is updated to current date and time minus time span value.
Retrieve Events between a specified START DATE/START TIME and Current Date/Time	Action: Select END DATE / END TIME value + ENTER Response: END DATE / END TIME value field is updated to current date and time. START DATE / START TIME value field is updated to current date and time minus time span value. Action: Select START DATE . Response: Entry port appears. Action: Enter start date such as 04 MAR 01 in entry port + ENTER. Response: New start date appears in START DATE / START TIME value field. Action: Select START TIME Response: Entry port appears. Action: Enter a value such as 14:00 (using 24-hour time format) into entry port + ENTER. Response: New start time appears in START DATE / START TIME value field.

Retrieve Events between a specified START DATE/START TIME and Current Date/Time continued	Action: Select START DATE Response: Entry port appears. Action: Enter start date such as 04 MAR 00 in entry port + ENTER Response: New start date appears in START DATE / START value field. Action: Select START TIME Response: Entry port appears. Action: Enter a value such as 06:00 (using 24-hour time format) into entry + ENTER. Response: New start time appears in START DATE / START TIME value field. Action: Select END DATE target. Response: Entry port appears. Action: Enter end date such as 04 NOV 00 in entry port + ENTER Response: New END date appears in END DATE / END TIME value field. Action: Select END TIME Response: Entry port appears. Action: Enter a value such as 06:00 (using the 24-hour time format) into the entry port + ENTER Response: End time appears in END DATE / END TIME value field.
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Process Alarms at Unit Level Retrieval	Action: Select PROCESS ALARMS target. Response: Next layer of choices appears: UNIT LEVEL, PRIMMOD LEVEL POINT LEVEL Action: Select UNIT LEVEL Response: Next level of choices appears: UNIT group and AREA UNITS plus CLR Action: Enter one or more units into UNIT group targets. NOTE: • Selection of AREA UNITS target will make UNIT targets inoperable. • Reselection of AREA UNITS target will deselect it and make UNIT targets operable again. Response: Unit numbers entered appear in individual Unit group. DISPLAY and PRINT become active. (Notice the cyan triangle in DISPLAY target.) NOTE: The printer of choice can be entered or changed at any time by entering printer number into PRTR target on Event History Retrievals display. Printer number entered remains the printer of choice until it is changed or station is reloaded. Action: Select DISPLAY or select PRINT Response: Data for PROCESS ALARMS journal for UNIT LEVEL is displayed or data for PROCESS ALARMS journal for UNIT LEVEL is printed to printer number of choice. NOTE: R600 provides the ability to return to the previous screen with all previous selections preserved by using the PRIOR DISP key.
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Process Alarms/Primmod Level Event Retrieval	Action: PROCESS ALARMS Response: Next layer of choices appears: UNIT LEVEL, PRIMMOD LEVEL POINT LEVEL. Action: Select PRIMMOD LEVEL Response: UNIT group, POINT group, AREA UNITS and CLR targets become available on screen. NOTE: DISPLAY and PRINT targets are unavailable until one or more units and one or more points are entered. Action: Enter one or more units into UNIT group targets. Response: Unit ID appears inside the unit group target. NOTE: If ALL units in the local area are desired, select the AREA UNITS target. Action: Enter one or more points into the POINT group targets. Response: Point information, as entered, appears in the point targets. DISPLAY and PRINT targets become available. Action: Select DISPLAY or select PRINT Response: Data for PROCESS ALARMS journal for PRIMMOD LEVEL is displayed or data for the PROCESS ALARMS journal for PRIMMOD LEVEL is printed to printer number of choice.
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Process Alarms (at the point level only) Retrieval	Action: Select PROCESS ALARMS Response: Next layer of choices appears UNIT LEVEL, PRIMMOD LEVEL, POINT LEVEL Action: Select POINT LEVEL Response: POINT group targets appear. NOTE: SEQUENCE OF EVENTS becomes inactive (dimmed) because SOEs are <u>unit related</u> only. Action: Enter one or more points into the POINT group targets. Response: Points appear in the POINT group targets. DISPLAY and PRINT targets become available. Action: Select DISPLAY or select PRINT Response: Data for PROCESS ALARMS journal for POINT LEVEL is displayed or data for PROCESS ALARMS journal for POINT LEVEL is printed to printer number entered previously. NOTE: For OPERATOR MESSAGES and PROCESS CHANGES perform same steps as described above for PROCESS ALARMS.
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Sequence of Events Retrieval	Action: Select the SEQUENCE OF EVENTS Response: AREA UNITS, UNIT groups plus CLR targets become visible. Action: Select AREA UNITS Response: UNIT group becomes inactive (dimmed). DISPLAY and PRINT targets become available. NOTE: If specific units are desired, deselect AREA UNITS Action: Select DISPLAY or select PRINT Response: SEQUENCE OF EVENTS data is displayed or SEQUENCE OF EVENTS data is printed to printer number of choice.
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System Events Retrieval	Action: Select SYSTEM MAINT target from the EHR display Response: CLR, NODE group and ALL NODES appear. Action: Enter one or more node numbers into NODE Response: Node numbers appear in targets. Action: Or, if <u>all</u> nodes are desired, select ALL NODES Response: CLR target disappears and NODE group is dimmed. NOTE: At any time, if a different selection is desired, either Clear or deselect target. Action: select DISPLAY target or select PRINT target Response: Data for System Maintenance for either individual nodes entered or nodes are displayed or data for System Maintenance for either individual nodes entered or ALL nodes is printed to printer number entered previously. NOTE: For SYSTEM ERRORS, SYSTEM STATUS, STATUS NOTIFY , and ENGR CHANGES follow the same steps outlined above for SYSTEM MAINT.
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**Process Network System
Status, BOX Level
Procedure**

Action: Select **PROCESS NETWORK SYSTEM STATUS**

Response:

ALL NTWKS and **NETWORK #** targets become visible.

ALL BOXES and **BOX** group are dimmed.

CLEAR BOXES is not visible.

Action: Select **NETWORK #** and enter Network number.

Response: Number appears in **NETWORK #**. **ALL BOXES**,
BOX group, and **CLEAR BOXES** become available.

Action: Enter one or more Box numbers in **BOX** group – or select
ALL BOXES

Selection of **ALL BOXES** implies all boxes included in the
network # previously entered.

Response: **BOX** numbers appear in targets. **DISPLAY** and
PRINT targets become available.

NOTE: To clear Box numbers already entered: first select the Box,
then **CLEAR BOXES** and press **ENTER** at the prompt,
or enter a new Box number after selection.

Action: Select **DISPLAY** or select **PRINT** target.

Response: Data for **PROCESS NETWORK SYSTEM STATUS**
journal at BOX level is displayed or data for **PROCESS**
NETWORK SYSTEM STATUS
journal at BOX level is printed to printer number entered
previously.

Process Network System Status, All Ntwks Level Retrieval	Action: Select PROCESS NETWORK SYSTEM Response: ALL NTWKS and NETWORK # become visible. ALL BOXES BOX group targets are dimmed. CLEAR BOXES is not visible. Action: Select ALL NTWKS NOTE: The selection of the ALL NTWKS target implies <u>all</u> <u>boxes</u> on <u>all</u> networks configured on your system. Response: NETWORK # now becomes inactive. DISPLAY and PRINT Action: Select DISPLAY or select PRINT target. Response: Data for PROCESS NETWORK SYSTEM STATUS, and ALL networks is displayed or data for PROCESS NETWORK SYSTEMSTATUS, and ALL networks is printed to printer number entered previously.
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Process Variable Retrieval

Call up, print, or display
the process variable
retrieval targets

Action: Select: |  |

Response: The following targets appear

Action: Select:  to select type of historical
values to retrieve

 +  type in a group
number, **or** tag
name(s) for up
to eight points

+ 

Response: The selected type of historical values
are retrieved

Action: Select: SELECT the START/STOP TIME/DATE
targets and for each, enter the values you
want.

Action: Select: 

Response: The variables are displayed on the screen

Action: Select:  Print
variables

Action: Type in printer number + 

Group Display Edit Function

The operator is allotted 10 groups (groups 391 through 400). These groups allow the operator to organize existing points in any order, but are not permanent, and can be changed at any time.

Change Group displays 	Action: Select:  Response: The following targets appear (391-400) Action: Select:  OR  Response: If you selected OTHER, a text box appears for you to enter a group number Action: Type in a group number 1 to 390+ 
Change Group titles	Action: Select:  Response: Targets appear allowing you to enter a new group title Action: Type in new group title +  Response: A new group title is now entered
Change Tag names	Action: Select:  Response: Targets appear allowing you to enter a new tag name in positions (1-8) Action: Type in new tag name +  Response: Tag name changes
Delete a point on the display	Action: Select:  Response: Targets appear allowing you to delete a new tag name in positions (1-8) Action: Select tag name +  Response: Point is now deleted

Overview Edit Display Function

Change Group element	Action: Select:  OVERVIEW EDIT DISPLAY Response: The following target appears Action: Select:  GROUP ELEMENT Response: The following targets appear  +  to delete (hide) the group from the display  +  to suppress alarm displays for the selected group  +  to enable all normal displays for the selected groups Response: Group element changes
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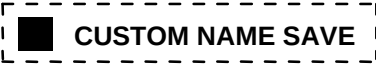
Removable Media Initialization

Initialize removable media	Action: Select:  REMOVABLE MEDIA INITIALIZATION Response: The following targets appear Action: Select:  OR  Select appropriate + drives Response: The following target appears Action: Select:  Type in 4 character volume name +  Response: The following target appears Action: Select:  Type in 20 character descriptor + 
----------------------------	--

Custom Name Save

Save custom names



Action: Select: 

Response: The following targets appear

Action: Select:

 OR  Select appropriate + drives

Response: The following target appears

Action: Select:  Type in 20 character descriptor + 

Response: The following target appears

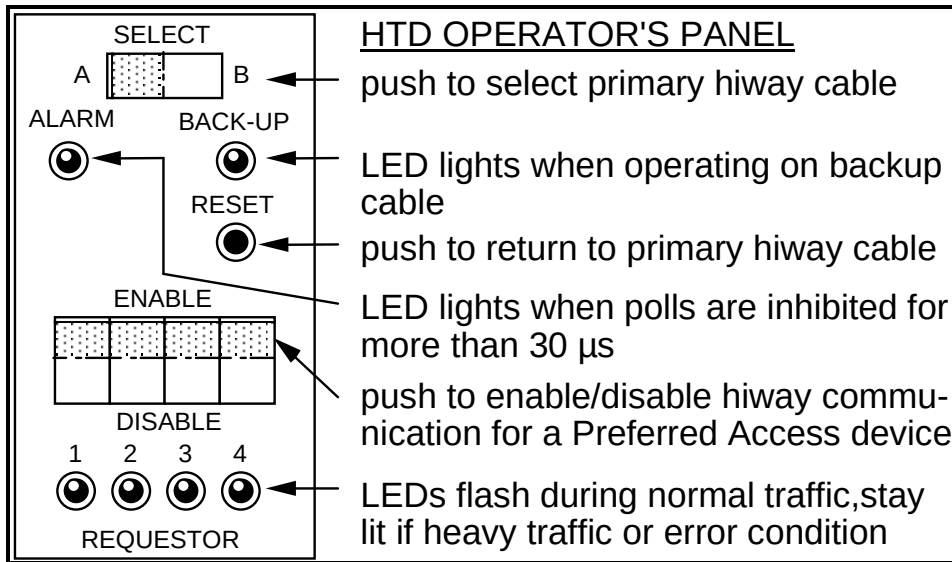
Action: Select: 

Response: Custom parameter names to the selected volume begins saving

Clear the Screen



HTD Panel Operations



SYSTEM STATUS COMMANDS

Call Up System Status Display

Call up System Status display	Action: Press:  Response: System Status display appears on the screen
-------------------------------	---

Change System Time and Date

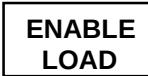
From the System Status display, perform the following.

Change System time and date	Action: Select:  Response: The prompt "ENTER TIME" appears with text box allowing you to type in time. Action: Type in Hours (00 - 23)  Minutes (00 - 59) +  Response: The prompt "ENTER DATE" appears with text box allowing you to type in date. Action: Type in Days (00 - 31)  Month (JAN, FEB, ...)  Year (01, 02,...) last two digits of the year +  Response: The time and date entry appears in the upper right corner of the display.
-----------------------------	---

Call Up System and Node Status Displays

Call up System and Module Status Displays	Action: Select:  Response: System Status display appears on the screen Action: Select one of the nodes object boxes, then select  Response: The selected node's corresponding node status display appears
---	---

Enable/Disable Node Auto Loads and Dumps

Enable/Disable auto loads and dumps	Action: Select: The node number of the desired node on the node status display. Response: Node number becomes highlighted Action: Select:  Response: Prompt: MAKE SELECTION Action: Select:     Response: Function is activated, current load state line shows change
-------------------------------------	--

Shut Down a Node

Shut down a node	Action: Select: The Node Number of the desired node on the node status display. Response: Node number becomes highlighted Action: Select:  +  Response: Node shuts down  Key may be required before node will shut down
------------------	--

Swap LCN Cables

Swap cables	Action: Select: CABLE A + ENTER Response: Cable B swaps to Cable A Action: Select: CABLE B + ENTER Response: Cable A swaps to Cable B
-------------	---

Load a NIM, HG, AM, HM, or CM

NOTE: The node must be in the "PWR_ON" or "QUALIF" state. Press **RESET** at the node or execute a **SHUTDOWN** to get the node in the proper state.

Load a NIM, HG, AM, HM, or CM	Action: Select node number on the node status display Response: The following target appears Action: Select: MAN LOAD Response: The following target appears Action: Select: OPERATOR PROGRAM Response: Targets appear to specify program load source Action: Select: DEFAULT SOURCE to load from the Network ALTERNATE SOURCE to load from removable media drive number must be specified + EXECUTE COMMAND Response: Targets appear to specify data load source Action: Select: DEFAULT SOURCE to load from the Network ALTERNATE SOURCE to load from removable media drive number must be specified + EXECUTE COMMAND Response: Node loads and status changes to OK
-------------------------------	--

Display Units Controlled by AM or CM

Display units controlled by AM or CM	Action: Select node number on the node status display Response: The following target appears Action: Select: UNIT CONTROL Response: Units controlled by AM or CM are displayed on the screen
---	---

Enable/Disable History Collection on an HM

Enable or disable History Module collection	Action: Select node number on the HM Status display Response: The following target appears Action: Select: HIST COLLECT Response: The following targets appear Action: Select: ENABLE COLLECT DISABLE COLLECT Response: Depending on target selected, History Module collection is enabled or disabled
--	---

Enable/Disable HG - Hiway Alarming

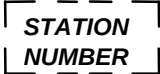
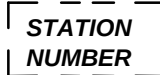
Enable or disable Hiway alarming	Action: Select node number on the Hiway Status display NODE NUMBER Response: The following target appears Action: Select: HWY ALRM ENB/DSB Response: The following targets appear Action: Select: ENABLE DISABLE Response: Depending on target selected, Hiway Alarming is enabled or disabled
---	--

CONSOLE STATUS COMMANDS

Call Up Console Status and Assignment Display

Call up Console Status display	Action: Press:  Response: Console Status display appears on the screen Action: Press:  +  +  Response: Remote Console Status display appears on the screen
--------------------------------	---

View Peripheral Status - View Maintenance Information

View Peripheral status	Action: Select:  +  Response: Peripheral status appears on the screen
View Maintenance Information	Action: Select:  +  Response: Maintenance information appears on the screen

Change Printer Assignment

Change printer assignment	Action: Select:  Response: Prompt ENTER ASSIGNMENT Action: Select:  and type assignment number Action: Select:  Response: Printer assignment is changed
---------------------------	---

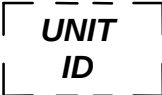
Display/Change Unit Status

Change Unit status

Action: Select

**UNIT
DISPLAY**

Response: Unit Assignment display is called up

Action: Select The image shows a screen with the text 'UNIT ID' in the center. The word 'UNIT' is on the top line and 'ID' is on the bottom line. They are both enclosed in dashed rectangular boxes.

Response: Unit is selected

Action: Select YES or NO to enable or disable operator control on the selected unit for any station loaded with the same area as this station.

UNIT ASSIGNMENT

YES

NO

OR

Action: Select one of the following targets:

System Alarm State:

ENABLE

to enable full alarming on the system alarm state

DISABLE

to disable alarming for the unit on all stations system-wide

INHIBIT

to inhibit alarming for the unit on all stations system-wide, including journaling

Console Alarm State:

ENABLE

to enable full alarming for stations in this console loaded with the same area as this station

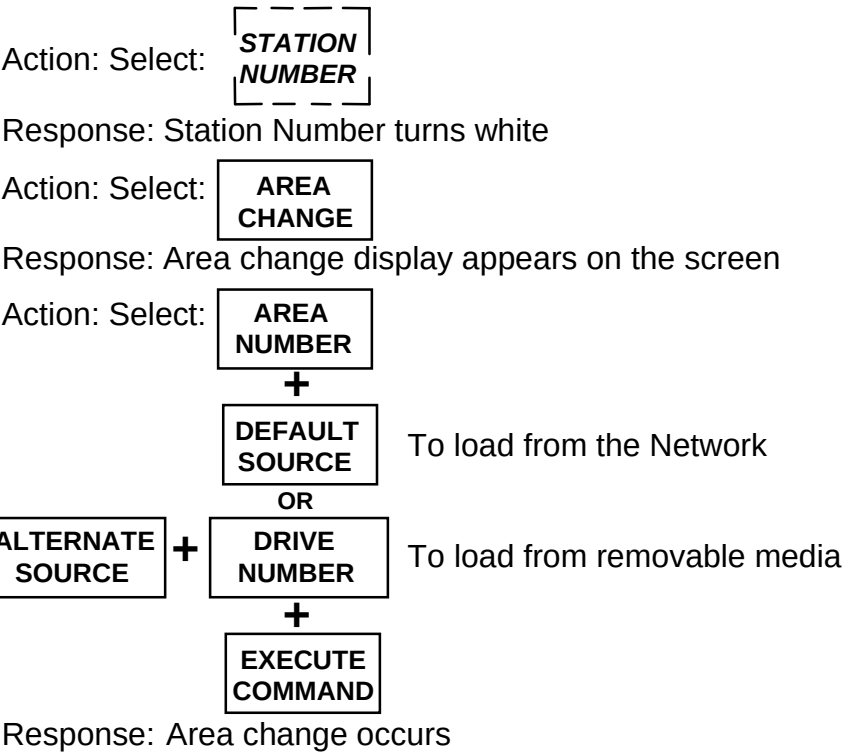
DISABLE

to disable alarm annunciation and displays for stations in this console loaded with the same area as this station

Response: Depending on target selected, different actions occur.

Area Change

Make an area change



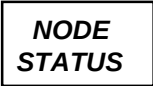
Reload Schematics and FFLs

Reload of memory resident Schematics and FFLs	Action: Select: RELOAD SCHM/FFL Response: The color of the target changes from green to white and a message, MAKE SELECTION appears at the top left corner of the local node. Three new targets NODExx, AREAYy, and CONSOLEzz appear with the text PASS and FAIL. (xx is the node number of the currently selected US, yy (upto 8 characters) is the area name, and zz is the console number.) To reload schematics and FFLs in a single node: Action: Select: NODExx Response: The color of the target changes from green to white and the message PRESS ENTER TO EXECUTE appears at the top left corner of the local node Action: Press ENTER Response: If the reloading is successful, the station node number is displayed next to PASS. If the reloading fails, the station node number is displayed next to FAIL NOTE: The selected node number(s) appears in the FAIL string, if • It is not in the running state. • It is loaded with R630 or prior release. • Reload, area change, or node load is in progress in the selected node. • Memory resident schematic is displayed in the selected node. To reload schematics and FFLs for all stations in a console that are in the same area: Action: Select: AREA yy Response: The color of the target changes from green to white and the message PRESS ENTER TO EXECUTE appears at the top left corner of the local node
--	--

	Action: Press ENTER Response: The node numbers next to PASS indicates that reload request is accepted in those nodes. The node numbers next to FAIL indicates that reload request is not accepted in those nodes To reload schematics and FFLs for all stations in a console regardless of the area: Action: Select: CONSOLE Response: The color of the target changes from green to white and the message PRESS ENTER TO EXECUTE appears at the top left corner of the local node Action: Press ENTER Response: The node numbers next to PASS indicates that reload request is accepted in those nodes. The node numbers next to FAIL indicates that reload request is not accepted in those nodes NOTE: More than one node number can be displayed separated by commas next to PASS or FAIL. For example: When nodes 5, 10, 15, 20 and 25 are in area yy or in console zz and if the user selects the Area yy or the Console zz target, based on the success or failure the status is shown as: PASS: 5, 10, 25 FAIL: 15, 20
--	---

HIWAY STATUS COMMANDS

Call up Hiway Status Display

Call up the Hiway Status display	Action: Select:  Response: The system status display appears Action: Select: NIM or HG node (for Hiway) Response: Network node becomes highlighted Action: Select:  Response: Process Networks Node Status Display appears Action: Select:  Response: Hiway Status display appears
---	---

Display Box Descriptors on Hiway Status Display

Display box descriptors at the Hiway Status display	Action: Select:  Response: Box descriptors appear on the left side of the screen
--	---

Load Trend Memory Program (Basic Controller)

Load Trend Memory program	Action: Select: TREND MEMORY Response: Text box appears Action: Type in box number or select ALL BOXES + ENTER Response: Targets appear to specify load source Action: Select: DEFAULT SOURCE to load from the Network ALTERNATE SOURCE to load from removable media, specify the drive number. + EXECUTE COMMAND OVERWRITE (optional) to overwrite the currently running program START FUNCTION to begin loading the program Response: Program begins loading
---------------------------	--

Swap Hiway Cables

Swap Hiway cables	Action: Select: HIWAY CMND Response: The following target appears Action: Select: SWAP HIWAY + EXECUTE COMMAND Response: Hiway cables swap
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Initialize Box Address

Initialize box address	Action: Select: HIWAY CMND Response: The following target appears Action: Select: INIT ADDR + EXECUTE COMMAND Response: Box address becomes initialized
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Enable/Disable all Hiway Swaps

Enable or disable all Hiway swaps	Action: Select: HIWAY CMND Response: The following targets appear Action: Select: ENABLE SWAP OR DISABL SWAP + EXECUTE COMMAND Response: Depending on target selection, all Hiway swaps are enabled or disabled
-----------------------------------	---

Enable/Disable Periodic Hiway Swaps

Enable or disable periodic Hiway swaps	Action: Select: HIWAY CMND Response: The following target appears Action: Select: ENABLE PERIOD OR DISABL PERIOD + EXECUTE COMMAND Response: Depending on target selection, periodic Hiway swaps are enabled or disabled
--	--

Change Hiway or Box Control State (without an HTD)

(Note: This is not an option when a Hiway Traffic Director (HTD) is included.)

Change Hiway or Box control state	Action: Select: HIWAY CMND OR BOX CMND Type in box number Response: The following targets appear Action: Select: BASIC CTRL READ CTRL FULL CTRL TEST CTRL + EXECUTE COMMAND Response: Depending on target selection, Hiway or Box control state changes
-----------------------------------	---

Enable/Disable Hiway Checkpointing

Enable or disable Hiway checkpointing	Action: Select: HIWAY CMND Response: The following targets appear Action: Select: CKPT ENABLE OR CKPT DISABL Response: Depending on target selection, Hiway checkpointing is enabled or disabled
---------------------------------------	---

Box Commands

Execute a box command	Action: Select: BOX CMND OR BOX STATUS Type in box number ENTER Response: The following targets appear Action: Select one of the following targets START to restart a CB, EC, MC, RCD, DHP, or PIU ENABLE PROC to change MC state from idle to processing, clears PIU or DHP report buffers, and enables processing. IDLE to change MC state from processing to idle ACK to clear PIU or DHP processing overload and A/D offset errors CONFIG CLEAR to clear the DHP database and reset the DHP + EXECUTE COMMAND Response: Depending on target selection, Box command is executed
-----------------------	--

Box Commands (RCD)

Execute an RCD Box command	Action: Select: BOX CMND OR BOX STATUS Type in box number Response: The following targets appear Action: Select one of the following targets <table border="1"> <tr> <td>START RC</td><td>to restart the reserve controller in a UAC system</td></tr> <tr> <td>SWBACK PRI</td><td>to cause the RCD to switch back to the primary</td></tr> <tr> <td>START PRI</td><td>to restart the primary controller in a UAC system</td></tr> <tr> <td>TEST PRI</td><td>to do a test backup of a primary controller in a UAC system</td></tr> <tr> <td>SAVE PRI</td><td>to force a backup of a primary controller in a UAC system</td></tr> <tr> <td>DISABL PRI</td><td>to disable a primary controller in a UAC system</td></tr> <tr> <td>ENABLE PRI</td><td>to enable a primary controller in a UAC system</td></tr> <tr> <td colspan="2" style="text-align: center;">+</td></tr> <tr> <td>EXECUTE COMMAND</td><td></td></tr> </table> Response: Depending on target selection, RCD Box command is executed	START RC	to restart the reserve controller in a UAC system	SWBACK PRI	to cause the RCD to switch back to the primary	START PRI	to restart the primary controller in a UAC system	TEST PRI	to do a test backup of a primary controller in a UAC system	SAVE PRI	to force a backup of a primary controller in a UAC system	DISABL PRI	to disable a primary controller in a UAC system	ENABLE PRI	to enable a primary controller in a UAC system	+		EXECUTE COMMAND	
START RC	to restart the reserve controller in a UAC system																		
SWBACK PRI	to cause the RCD to switch back to the primary																		
START PRI	to restart the primary controller in a UAC system																		
TEST PRI	to do a test backup of a primary controller in a UAC system																		
SAVE PRI	to force a backup of a primary controller in a UAC system																		
DISABL PRI	to disable a primary controller in a UAC system																		
ENABLE PRI	to enable a primary controller in a UAC system																		
+																			
EXECUTE COMMAND																			

UCN STATUS COMMANDS

Call up UCN Status Display

Call up the UCN
Status display

Action: Select: 

Response: The system status display appears

Action: Select: NIM node (For UCN).

Response: NIM box becomes highlighted

Action: Select: 

Response: UCN Status Display appears

OR

Action: Select: 

Response: The system status display appears

Action: Select: NIM node (For UCN).

Response: NIM box becomes highlighted

Action: Select: 

Response: Process Networks Node Status Display

Action: Select the network you want, then select



Response: UCN Status Display appears

Change UCN or XPM Control State

Change UCN control states	Action: Select: CONTROL STATES from the UCN Status display Response: The following targets appear Action: Select: UCN FULL CONTROL OR UCN BASIC CONTROL + ENTER Response: UCN control state changes to full or basic
Change XPM control states	Action: Select: CONTROL STATES from the UCN Status display Response: The following targets appear Action: Select: DEV FULL CONTROL OR DEV BASIC CONTROL Response: The following targets appear Action: Select: Specific XPM OR ALL DEVICES + ENTER Response: XPM control state changes to full or basic

Enable/Inhibit Automatic Checkpointing

Enable or inhibit UCN and device automatic checkpointing	Action: Select: AUTO CHECKPT from the UCN Status display Response: The following target appears Action: Select: UCN CKPT ENABLE OR UCN CKPT INHIBIT + ENTER Response: UCN checkpointing is enabled or inhibited Action: Select: DEV CKPT ENABLE OR DEV CKPT INHIBIT + ENTER Response: UCN device checkpointing is enabled or inhibited Action: Select: CONTROL STATES Response: The following targets appear Action: Select: DEV FULL CONTROL OR DEV BASIC CONTROL Response: The following targets appear Action: Select: Specific XPM OR ALL DEVICES + ENTER Response: UCN and device auto checkpointing is enabled or inhibited
--	--

Select UCN Cable

Select UCN cables	Action: Select: UCN CABLE STATUS from the UCN Status display Response: The following targets appear Action: Select: SELECT CABLE A OR SELECT CABLE B Response: The following targets appear Action: Select: ALL NODES OR Specific XPM + ENTER [Note warning on screen] Response: Cable selection changes for all nodes or a specific XPM Note: With TPN R684, two new FTE status schematics are introduced to aid in the diagnostics and troubleshooting of the FTE cable problems or for viewing the FTE status. \$FTESTS1 shows the FTE status of the UCN nodes assigned to the selected ENIM. \$FTESTS2 shows the status of all the FTE devices in the FTE Community. For more information about the \$FTESTS1 and \$FTESTS2 displays, refer to <i>Process Operations Manual</i>.
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Enable/Disable Automatic UCN Cable Swaps

Enable or disable automatic UCN cable swaps	Action: Select: UCN CABLE STATUS from the UCN Status display Response: The following targets appear Action: Select: ALL NODES OR Specific XPM Response: The following targets appear Action: Select: ENABLE SWAP OR DISABLE SWAP + ENTER Response: Automatic cable swapping is enabled or disabled
---	--

Idle an XPM

Idle an XPM	Action: Select: RUN STATES + Specific XPM from the UCN Status display Response: The following targets appear Action: Select: IDLE + ENTER Response: The XPM is idled
-------------	--

Start up an XPM

Start up an XPM	Action: Select: RUN STATES + Specific XPM from the UCN Status display Response: The following target appears Action: Select: STARTUP Response: The following targets appear Action: Select: WARM STARTUP OR COLD STARTUP + ENTER Response: The XPM is started
-----------------	---

Idle or Shut Down an XPMM, or I/O Processor

Shut down or idle an XPMM, or I/O Processor	Action: Select: Specific XPMM + DETAIL STATUS from the UCN Status display Response: The detail display for a specific XPM appears on the screen Action: Select: RUN STATES + Specific Device Response: The following targets appear Action: Select: IDLE OR SHUTDOWN + ENTER Response: The XPM or I/O is shut down or becomes idled NOTE: <i>The XPM must be in IDLE before it can be shut down</i>
---	---

Swap XPMMs

Swap XPMMs	Action: Select: Specific XPMM + DETAIL STATUS from the UCN Status display Response: The detail display for a specific XPM appears on the screen Action: Select: RUN STATES + XPMM Response: The following targets appear Action: Select: SWAP PRIMARY + ENTER Response: The XPMMs swap
------------	--

DATABASE SAVE (CHECKPOINT) FUNCTIONS

Save AM or CM Control Database

Save AM or CM
control database

Action: Select:



Response: The system status display appears.

Action: Select desired AM or CM node.

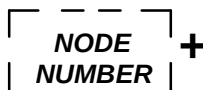
Response: Node object box becomes highlighted.

Action: Select:



Response: The application or communication module display appears on the screen.

Action: Select:



+



Response: The following targets appear

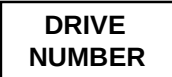
Action: Select:



To load from the Network



+



+



Response: The database for the AM or CM is saved

Enable/Disable Auto Save for NIM, HG, AM, or CM

Enable or
disable auto save for
the
NIM, HG, AM,
or CM

Action: Select:



Response: The system status display appears

Action: Select desired NIM, HG, AM or CM Node

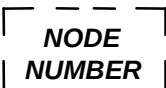
Response: Node box becomes highlighted

Action: Select:



Response: The appropriate network display
appears on the screen

Action: Select:



+

AUTO
SAVE

Response: The following targets appear

Action: Select:



OR

DISABLE
SAVE

Response: Auto save is enabled or disabled for the NIM,
AM, HG, or CM

Save or Load Box Database

Save or load
a box
database

Action: Select:



Response: The System Status display appears

Action: Select desired HG node

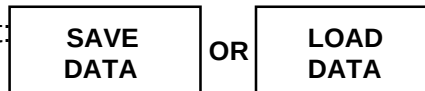
Response: Node box becomes highlighted

Action: Select:



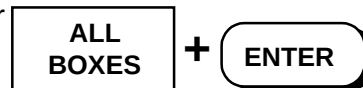
Response: Hiway status display appears on the screen

Action: Select:



Response: The following target appears

Action: Type in box number
or select



Response: The following targets appear

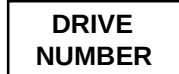
Action: Select:



To load from the Network



+



To load from
removable media

+



Response: The box database is loaded or saved

Save or Load NIM/XPM/LM Database

Save or load
NIM/XPM/LM
database

Action: Select:



Response: The system status display appears.

Action: Select NIM node number (for UCN node)

Response: NIM Node box becomes highlighted

Action: Select:



Response: UCN Status Display appears.

Action: Select:



Response: The following targets appear

Action: Select one of the following targets

PROGRAM LOAD	to load XPM program + data (XPMM "ALIVE")
RESTORE DATA	to restore XPM & NIM data (XPMM "IDLE")
SAVE DATA	to checkpoint XPM & NIM data
LADDER LOAD	to load ladder logic (LM)
LADDER SAVE	to checkpoint ladder logic (LM)

Response: The following targets appear

Action: Select:



to load from the Network



+



to load from
removable media

+



Response: Depending on target selection, data is
loaded or saved

SEQUENCE OPERATIONS

Process Module Group

Call up procedure	Action: Press:  Response: Prompt: ENTER GROUP NUMBER Action:  +  └ type in group number 401 to 450 Response: Selected process module group appears on the screen
Call up procedure from the Module Summary Display	Action: Select:  Response: Selected process module group appears on the screen

Process Module Detail

Call up procedure	Action: Press:  Response: Prompt: ENTER POINT ID Action:  +  └ Type in tagname of process module Response: Selected process module detail display appears on the screen
Call up procedure from the Module Summary Display	Action: Select:  Response: Selected process module detail display appears on the screen

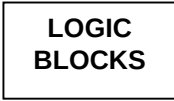
Timers

Call up a Timer	Action: Select:  OR  (MC Display) (XPM Display) Response: Timer display appears on the screen
Start, stop, or reset a Timer	Action: Select:  Response: The following targets appear allowing you to change a Timer Action: Select:  to start the timer  to stop the timer  to reset the timer Response: Prompt PRESS ENTER TO EXECUTE Action: Press:  Response: Timer changes to specific target selection
Preset a Timer	Action: Select:  Response: The following target appears Action: Select:  Response: Prompt VALUE? with text box appears allowing you to enter a new value Action: Type in new value +  Response: Timer is now preset
Change time base	Action: Select:  Response: Prompt PRESS ENTER TO EXECUTE Action: Press:  Response: Time base changes (min. to sec. or sec. to min.)

Numerics

Call up from a Process Module Detail Display	Action: Select:  OR  (MC Display) OR  (XPM Display) Response: Numeric display appears on the screen
Change a numeric value	Action: Select:  Response: Box appears to enter a numeric value Action: Type in Real number Response: Numeric value updates to show real value

Logic Blocks

Call up from the Module Summary Display	Action: Select:  Response: Logic Block display appears on the screen Note: Logic Block are associated with MC and XPM devices only.
---	--

Flags

Call up from a Process Module Detail Display	Action: Select:  OR  (MC Display) OR  (XPM Display) Response: Numeric display appears on the screen Action: Select:  Response: Numeric becomes highlighted in white Action: Press:  OR  +  Response: Flag state changes
--	--

Load a Sequence Program

Loading procedure from a Process Module Display	Action: Select: LOAD SAVE Response: Load target appears on the screen
Loading from a History Module	Action: Select: LOAD Response: DEFAULT SOURCE OR ALTERNATE SOURCE Action: Select: DEFAULT SOURCE + EXECUTE COMMAND Response: Process Module detail display with sequence name appears on the screen Action: Select: SEQUENCE NAME + START Response: Loading begins
Loading from an Alternate Source	Action: Select: LOAD Response: DEFAULT SOURCE OR ALTERNATE SOURCE Action: Select: ALTERNATE SOURCE Response: Display appears to make appropriate drive selections Action: Select drive 1, 2, 3, etc. depending on which one has the removable media Response: Process Module detail display appears on the screen Action: Select: SEQUENCE NAME + START Response: Loading begins

Save a Sequence Program

Saving from a Process Module Display	Action: Select: LOAD SAVE Response: Save target appears on the
Saving using a History Module	Action: Select: SAVE Response: DEFAULT SOURCE OR ALTERNATE SOURCE Action: Select: DEFAULT SOURCE + EXECUTE COMMAND Response: Process Module detail display with sequence name appears on the screen Action: Select: START Response: Saving begins
Saving using an Alternate Source	Action: Select: SAVE Response: DEFAULT SOURCE OR ALTERNATE SOURCE Action: Select: ALTERNATE SOURCE Response: Display appears to make appropriate drive selections Action: Select drive 1, 2, 3, etc. depending on which one has the removable media Response: Process Module detail display appears on the screen Action: Select: START Response: Saving begins

Change Module Operating State

Call up from a Process
Module Group or Detail
Display

Action: Select:  Under OPER column

Response: The following targets appear allowing you to change a Process Module Operating state

Action: Select:

READY

to put sequence in READY state (MC, PM, & APM)

STRT

to start sequence

STOP

to stop sequence

OFF

to set module state to OFF

EMSD

to initiate the Emergency Shutdown Handler (if enabled)

**SHUT
DOWN**

to initiate the Shutdown Handler (if enabled)

HOLD

to initiate the Abnormal Condition Handler (if enabled)

RESTART

to restart the sequence when in a HOLD or SHDN state (MC, XPM)

Response: Process Module Operating State will change once enter is selected

Action: Select:

ENTER

OR

ENTER

Response: Process Module Operating state changes

Change Sequence Execution State

Change the Execution State from a Process Module Group or Detail Display	Action: Select:  Under EXEC column Response: The following targets appear allowing you to change the Execution state Action: Select:  to start the sequence  to stop the sequence Response: Execution State will change only if sequence program is loaded and ENTER is selected Action: Select:  OR  Response: Execution State changes
---	---

Change Sequence Execution Mode

Change the Sequence Mode from a Process Module Group or Detail Display	Action: Select:  under MODE column Response: The following targets appear allowing you to change the Execution Mode Action: Select:  to put the sequence in AUTO mode  to put the sequence in SEMI-AUTO mode  to put the sequence in SINGLE-STEP mode Response: Execution mode will change once enter is selected Action: Select:  OR  Response: Execution Mode changes
---	---

Select Sequence Override

Select Sequence Override

Action: Select one of the following targets

PHASE
NAME

under PHASE column

STEP
NAME

under STEP column

STATEMENT
NUMBER

Statement Number near top
center of Process Module
Detail Display

Response: The following targets appear allowing you to change the Phase/Step/Statement

Action: Select:

FWD

to go to next Phase/Step/
Statement

BACK

to go to previous Phase/Step/
Statement

Response: Phase/Step/Statement changes forwards or backwards once enter is selected

Action: Select:

ENTER

OR

ENTER

Response: Phase/Step/Statement changes

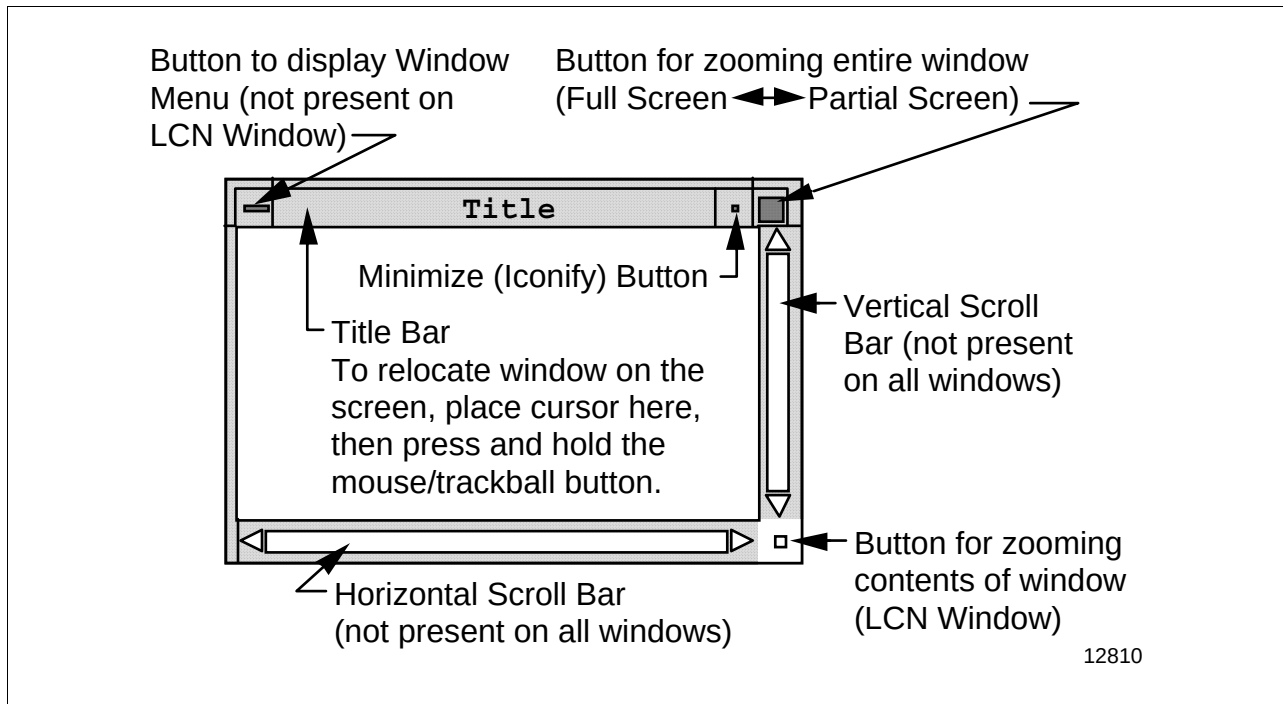
NOTE: Sequence Override can be performed for the MC only if the Execution State is ERxx/FAxx, or the Operating State is READY; for the XPM, the Execution State must be Exxx/Fxxx. Failure code F165 indicates that the operator has stopped the sequence program.

UNIVERSAL STATIONX WINDOWS

This procedure applies only to a Universal Station with X Windows.

To resize a window, place the cursor at an edge or corner of the window. The cursor will change to a small line and arrow that indicates the border(s) that will move. Press the mouse/trackball button and hold while dragging the border(s) to the desired location. Release the button.

The following illustration shows the cursor targets that can be used to manipulate the windows on the screen of a Universal Station^X.



To restore an iconified window:

- Double-Click with the cursor on the icon, or
- Click once with the cursor on the icon, and make the appropriate selection from the resulting menu.

To access items under the workspace (root) menu:

- Open the menu by placing the cursor in a screen area outside all windows, then pressing and holding down the mouse/trackball select button.
- Drag the cursor down the menu list to the desired selection, releasing the button to select that item. If this opens a submenu, click on the desired submenu item.

RULA DISCONNECT PROCEDURES

Use either of the following procedures to release a station from a RULA connection and regain use of the station for normal operations. Disconnect may be done on either the Host or Remote RULA Station. You cannot use the Quick Disconnect Procedure to disconnect the RULA diagnostic mode. For additional information, see the RULA User Manual.

Quick RULA Disconnect Procedure

Procedure for Quick RULA Disconnect	Action: On a Universal Station Engineering Keyboard, hold down CTL and press 5 on numeric keypad OR Action: On a Universal Station ^X Auxiliary Keyboard, hold down Ctl and press — on numeric keypad Response: Station is released from the RULA connection and the disconnection is journaled
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RULA Disconnect Procedure

Place keyswitch in the Engineer position.

Procedure for RULA Disconnect	Action: From the Console Status Display, select RULA CONTROL Response: RULA Control Display appears Action: Select Disconnect RULA Response: The station is released from the RULA connection and the disconnection is journaled
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FINDING POINT INTERCONNECTIONS

With TPN R660, Finding Point Interconnections is an optional feature added to a Universal Station that can search the system database within standard operating displays for connection references to a selected point. Although it uses the Find Names function, only a small amount of the Find Names options are available, and the operator does not see the Find Names interfaces.

Call up \$INTRCON schematic

Navigate from group display	Action: Select a point name and Press: SCHEM Response: Prompt: ENTER SCHEMATIC NAME Action: a # + ENTER type in \$INTRCON Response: The \$INTRCON page appears on the screen with the entity name as the selected point name																		
OR																			
Navigate from detail display	Action: Select: FIRST PAGE on Detail Display Response: The following targets may appear: <table><tr><td>FIRST</td><td>CTL ALGO</td><td>CTLCONCT</td><td>CONFIG</td><td>CL BLOCK</td><td>CUSTOM</td></tr><tr><td>PRIOR</td><td>GROUP</td><td>GRPASSOC</td><td>GRPHELP</td><td>ASSOC</td><td>CDETAIL</td></tr><tr><td colspan="6">\$INTRCON</td></tr></table> Action: Select: \$INTRCON Response: The \$INTRCON schematic appears on the screen with the entity name as the selected point name	FIRST	CTL ALGO	CTLCONCT	CONFIG	CL BLOCK	CUSTOM	PRIOR	GROUP	GRPASSOC	GRPHELP	ASSOC	CDETAIL	\$INTRCON					
FIRST	CTL ALGO	CTLCONCT	CONFIG	CL BLOCK	CUSTOM														
PRIOR	GROUP	GRPASSOC	GRPHELP	ASSOC	CDETAIL														
\$INTRCON																			

Start/Stop Operator Find Names from \$INTRCON Schematic

Make entries in \$INTRCON schematic	Action: On the \$INTRCON schematic, enter a descriptor that can be up to 64 characters long in the FNM Query entry port and press ENTER. The descriptor is the name of the Documentation Tool FNM Query that was previously established by the engineer Note: The descriptor defines the Documentation Tool query that contains the point interconnection search limits. Action: Select PRT target and press ENTER to have the point interconnection search results printed on the printer. A PRT value of ON indicates that output results go to the printer device number assigned by the engineer in the FNM query. A value of OFF indicates that the output results will not be printed. Action: Select FILE target and press ENTER to have the point interconnection search results sent to a file. A FILE value of ON indicates that the output results are to be sent to the file name assigned by the engineer in the FNM query. A value of OFF indicates that the output results are not sent to the file. Action: Select Default target and press ENTER to search the values specified in default parameters \$PICQRYA (for an AM point), \$PICQRYH (for a HG point), or \$PICQRYU (for a UCN point). When the value of Default is ON, it overrides any FNM Query descriptor entry Again, select the Default target and press ENTER so that it shows a value of OFF, which allows the use of the FNM query descriptor instead of the default queries. Only the engineer can make entries in the Documentation Tool FNM Query to define the descriptor files. The engineer can then develop a schematic that contains a list of descriptors for the operator. The engineer can also enter values into the \$PICQRYA, \$PICQRYU, \$PICQRYH parameters if the engineer wants the operator to use default parameters. All these features are optional targets and each requires knowledge of how the engineer has configured the system database, and the FNM queries used with \$INTRCON requests.
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Start the Search	Action: Select Request target and press ENTER Response: Starts the point interconnection search and sets Status to BUSY. As point interconnections are found, they are displayed on the \$INTRCON schematic. When the search is complete, Request is set to OFF and Status is set to INACT (Inactive). A cancelled request also sets Request to OFF and Status to INACT. NOTE: If the Status is UNAVAILABLE, this indicates that the engineer has not loaded the respective external load module for the point interconnection function into the station. In addition, any request for action produces an error response. Ask the engineer to load the external load module and prepare the station so that you can perform the point interconnection search function.
Monitor Search Results	When the search is in progress, you can monitor the search results by viewing the \$INTRCON schematic. The results are displayed below the Entity name with a reference to the Param name and the CL Block/Sequence name. A maximum of 72 values can be displayed on the results screen. When Request is set to ON and the other optional \$INTRCON parameters are set to default values, a search of all HM NET AM/UCN/HG checkpoints is initiated. NOTE: On systems with large databases, the request may take too long to complete. The operator can use the Universal Station for other operator functions during the search and return to the point interconnection search later to see the results.
Cancel the search	Action: Select Cancel target and press ENTER Response: The search request is cancelled. The operator can stop a Request that is in progress at any time. The Cancel target entry sets the Request value to OFF and the Status to INACT with an internal error showing as aborted. The rest of the results remain on the screen until the next request.

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